NATO COMEDS
Medical Naval Panel (MedNP)

Team 5: TCCC in the Maritime Sub Panel

Maritime Applications of TCCC Scenarios Deck

Updated: 24 JUN 2020

Maritime Applications of TCCC Scenario Development Assignments

	Scenario	Injuries	Assigned Nation
1.	**Missile Strike**	1. Burns – Skin 2. Blunt trauma 3. Penetrating injuries 4. Smoke inhalation 5. BLAST injuries	USA
2.	**Mine Torpedo**	1. Blunt Trauma 2. Penetrating injuries 3. Submersion / near drowning	NLD
3.	**Helicopter (Helo) Crash**	1. Fractures 2. Burns 3. Spine injuries	FRA
4,	**Collision**	1. Submersion / near drowning 2. Blunt Trauma 3. Hypothermia	USA
5.	**Ship Boarding**	1. Penetrating trauma 2. Blunt trauma from fall	CAN
6.	**Pick up From Water**	1. Immersion & submersion: near drowning and hypothermia 2. Chemical burns 3. Dehydration	DEU
7.	**FAC / FIAC**	1. Penetrating trauma	NOR

Learning Objective

APPLY your knowledge of TCCC to maritime scenarios

TCCC within the Maritime Environment

Tactical Combat Casualty Care (TCCC) guidelines should also be considered in the context of the maritime environment, which is unique when compared to land operations, for which there are nuances in application as well as other considerations. Although TCCC guidelines were developed for combat, there is the potential whereby such skill-sets could be applicable to non-combat scenarios. These scenarios might include entities such as collision at sea, fire or explosion, and helo-crash on deck. Combat tactical threats at sea can be varied, to include enemy fire from surface and sub-surface assets, asymmetric threats from small boats and aircraft, and attack while alongside. In addition, naval boarding parties may encounter direct enemy contact while conducting operations. TCCC is the standard of care in Prehospital Battlefield Medicine. Depending on a nation's approval authorities, there may be circumstances whereby such interventions may be authorized for non-combat scenarios.

Maritime Applications of TCCC

The naval environment is unique in several ways, which often includes remote areas of operation, which impacts time and space considerations and the ability to medevac. Additionally, each ship is unique in its role, size, vessel lay-out, and number of crew and configuration of the Damage Control/Casualty Clearing Organizations. In addition, shipboard injuries are unique with a preponderance of burns, inhalation injuries, and near-drowning.

Most navy ships have dedicated, integral medical personnel supported by a non-medical Casualty Clearing Team trained in applying advanced first aid and carrying stretchers throughout the ship (including through hatches and ladders). Depending on the mission, the medical team may be augmented by a surgical team or other medical capabilities. However, on smaller ships, there may not be many integral medical personnel, and medical support may be provided by Advanced First Aider(s), who are often a non-combat trade, such as cook, supply tech, or steward.

Knowing and understanding the characteristics and specific medical capabilities of your ship is vital in the maritime environment, and should be considered in relation to casualty management. It is imperative, therefore, to become familiar with the ship within the first 24 hours you are on board. By doing so, one will be better prepared to deal with emergency situations, to include casualty management. The structure/organization of each ship dictates a framework within which medical procedures must be performed, including the need to relocate medical and damage control stations depending on the casualty. Such preparation has potential to save one's life and that of others!

A final, important distinction regarding maritime scenarios is the unique perspective that saving the ship is the primary focus for all crew members. This focus may impact the priority that injured personnel would otherwise receive.

Recommended Triage of Care

Phase 1

Care Under Threat at Sea (CUT)

Care under threat at sea (care rendered by any sailor at the scene of injury while still under threat)

Phase 2

Tactical Field Care (TFC)

Basic management plan for Tactical Maritime Care (care rendered by any sailor while they are no longer under threat)

Phase 3

Care w/in Medical Space

Tactical medical care at sea (care rendered once the casualty has been moved to a medical space)

Recommended Maritime Phase Equivalents

- Phase 1 – Care under threat at sea

  Phase 1 occurs at the point of injury

  – Ship/Damage control is the first priority
  – Apply personal safety gear. Stay on stations; if not possible escape from threats
  – Direct casualty to move/move casualty to safe zone and to apply self-aid (if able)
  – Prevent additional injuries, hypothermia, and smoke inhalation
  – Extricate casualties from hazardous spaces
  – Stop life-threatening external hemorrhage, if feasible
    - Apply CoTCCC - recommended limb tourniquet for hemorrhage
  – Airway management is generally best deferred to Phase 2

- Move to Phase 2 – Basic Management Plan

Phase 1

Phase 2

Recommended Maritime Phase Equivalents

- Phase 2 – Basic Management Plan

  Phase 2 begins at the first secure area. The goal is to be outside the area of threat

  – Ensure safe casualty collection point and maintain tactical situational awareness
  – Casualties with an altered mental status should have weapons and communications equipment taken away immediately
  – Massive Hemorrhage
  – Airway Management and Breathing
  – Circulation
  – Inspect and dress wounds and check for additional wounds
  – Burns
  – Split fractures and re-check pulses
  – Drowning / near drowning
  – Assess for hypothermia
  – Communicate – with casualty (if possible) and with casualty clearing team/ ship medical authority (as soon as possible and as frequently as needed)
  – Cardiopulmonary resuscitation (CPR) = Phase 2 (CPR should be performed on all hypothermic/ near drowning patients)
  – Initiate transfer of casualty by the casualty-clearing teams to ship sick-bay or casualty-clearing station

- Move to Phase 3

Phase 2

Phase 3

Recommended Maritime Phase Equivalents

- Phase 3 – Care w/in Medical Space at Sea

  Phase 3: Move from location of Phase 2 to a medical treatment area on the ship (if available) or designated area

  – Re-assessment by repeating steps of MARCH-PAW and carry out necessary interventions, VS monitoring.
  – Hypothermia prevention
  – Penetrating eye trauma
  – Pain relief
  – Antibiotics: recommended for all open combat wounds
  – Triage casualties
  – Documentation of care
  – Prepare for evacuation

Phase 3

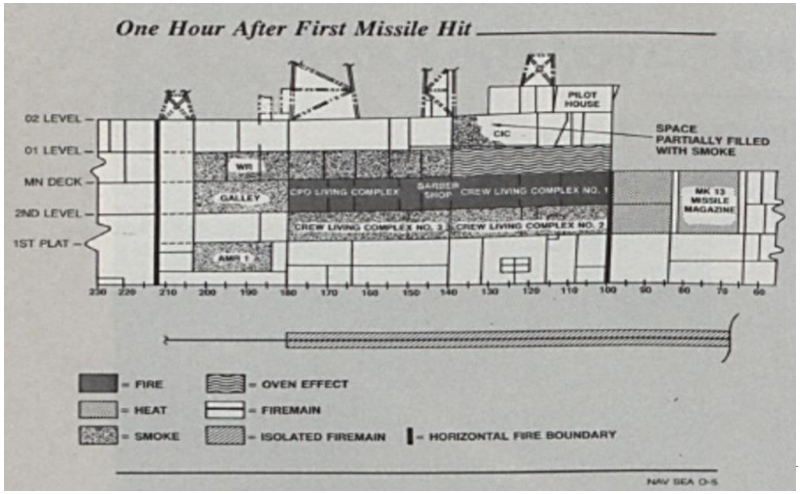
Scenario 1: Missile Strike at Sea
Case Background

- **Scenario is based off the USS STARK incident involving Exocet missile strike by Iraqi F-1 Mirage**
- **29 KIA, 8 DOW later, 21 injured**

Image from Google.com search.

TCCC in the Maritime Scenario 1

- USS STARK is conducting routine daytime patrol maneuvering off the coast of Saudi Arabia
- Exocet #1 punches through the USS STARK's hull near the port bridge wing, eight feet above the waterline-no detonation occurs but rocket fuel burns. Exocet #2 hits five forward of the first and detonates as designed
- Multiple sailors killed, wounded. Temperatures inside affected spaces are extremely high
- Single patient scenario described here for teaching purposes
- Medical was severely damaged. Triage area was the hangar bay
- A Navy Independent Duty Corpsman was injured



Images from Google.com search

Response Considerations

- What types of injuries anticipated?
- What is size & composition of medical response team? Level of training?
- What resources and supplies available at nearest Battle Dressing Station? What does the team carry?
- What is the pre-planned patient movement route? Alternate "safe" routes according to damage control team?
- Status of blood supply? Walking blood bank?
- Proximity to next echelon of care? Patient movement assets?
- Who handles the En route care?

TCCC in the Maritime Scenario 1

Phase 1: Care Under Threat at Sea (CUT)

Patient Injuries – 24 yo Male

- Right thigh with obvious bleeding-penetrating/shrapnel wound suspected
- Right chest and neck penetrating wounds
- Right upper back with suspected exit wound

Space has minimal flooding, extremely high temperatures

TCCC in the Maritime Scenario 1

Phase 1: Care Under Threat at Sea (CUT)

Patient - CUT Critical Actions

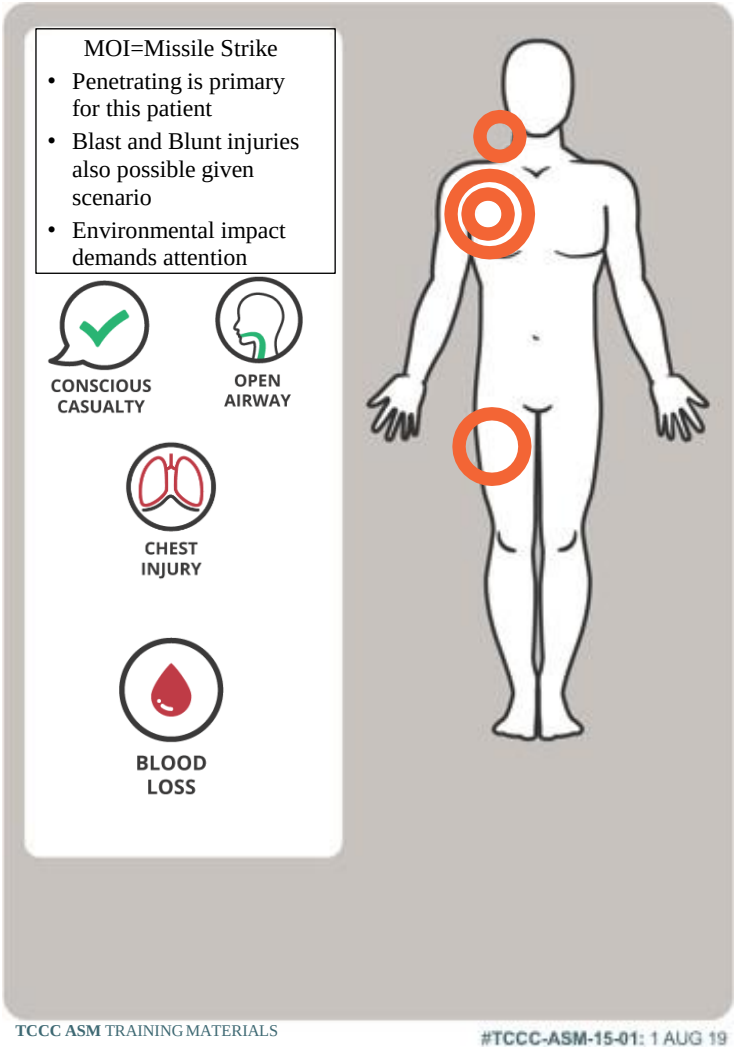
- Check scene safety
- Assess for Massive Hemorrhage & apply tourniquet
- Remove patient from compartment to safe space
- Address hypothermia

TCCC in the Maritime Scenario 1

Phase 2: Tactical Field Care (TFC) - Exam

What You See/Feel:

- Alert, speaks clearly, shivering
- Obvious bleeding about RT thigh, RT anterior chest, RT side of neck
- Weak, thread, rapid pulse
- Labored breathing, poor RT side chest rise



TCCC in the Maritime Scenario 1

Phase 2: Tactical Field Care (TFC)

Patient - TFC Critical Actions

- M:
  - Right Thigh Tourniquet
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression, chest seal placement-anterior and posterior chest wounds
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Pressure dressing to neck wound
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

TCCC in the Maritime Scenario 1

Phase 2->3

Patient Movement / En Route Care

Pt Movement/En Route Care Critical Actions

- Patient packaging
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Cervical-spine precautions
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

TCCC in the Maritime Scenario 1

End of Scenario 1

Scenario 2: Mine/Torpedo
Case Background

- **Scenario is based off the USS PRINCETON patrolling in the Persian Gulf**
- **The morning of 18 February 1991**
- **At 7:15 AM, two MN103 Manta bottom-mounted influence mines detonated**
- **Three crew members were injured, one seriously**

Image from Google.com search

TCCC in the Maritime Scenario 2

- USS PRINCETON is conducting routine daytime patrol
- On the morning of 18 February 1991, during Operation Desert Storm, *PRINCETON* was patrolling 28 nautical miles (52 km) off Failaka Island in the Persian Gulf
- At 7:15 AM, two MN103 Manta bottom-mounted influence mines detonated, one just under the port rudder and the other just forward of the starboard bow, the second explosion most likely being a sympathetic detonation caused by the first
- The blasts cracked the superstructure, buckled three lines in the hull, jammed the port rudder, flooded the #3 switchboard room through chilled water pipe cracks, and damaged the starboard propeller shaft
- Three crewmembers were injured, one seriously. Despite the severe damage, the forward weapons and the AEGIS combat system were back online within 15 minutes

TCCC in the Maritime Scenario 2

*A hull crack caused by the mine that detonated underneath PRINCETON

Images from Google.com search

Response Considerations

- What types of injuries anticipated?
- What is size & composition of medical response team? Level of training?
- What resources and supplies available at nearest Battle Dressing Station? What does the team carry?
- What is the pre-planned patient movement route? Alternate "safe" routes according to damage control team?
- Status of blood supply? Walking blood bank?
- Proximity to next echelon of care? Patient movement assets?
- Who handles the En route care?

Scenario 2a - Patient Injuries:

- **Patient 1:** 18 yo Male: Conscious, in water above the knee
- **Patient 2**: 29 yo Male: Conscious, in water above the knee, pulsatile bleeding from his wrist
- **Patient 3**: 37 yo Male: Unconscious, lying face down in water
- **Patient 4**: 25 yo Female: Conscious, bleeding from the face

Worsening flooding in the compartment

TCCC in the Maritime Scenario 2

Phase 1: Care Under Threat at Sea (CUT)

Patient - CUT Critical Actions

- Check scene safety
- Assess for Massive Hemorrhage & apply tourniquet
- Remove patient from water-filled compartment
- Address hypothermia

TCCC in the Maritime Scenario 2

- Triage your 4 patients
- Where do you start?
- Global sorting considerations:
  - Bleeding?
  - Walking?
  - Breathing?
  - Unconscious?
  - Pulses?

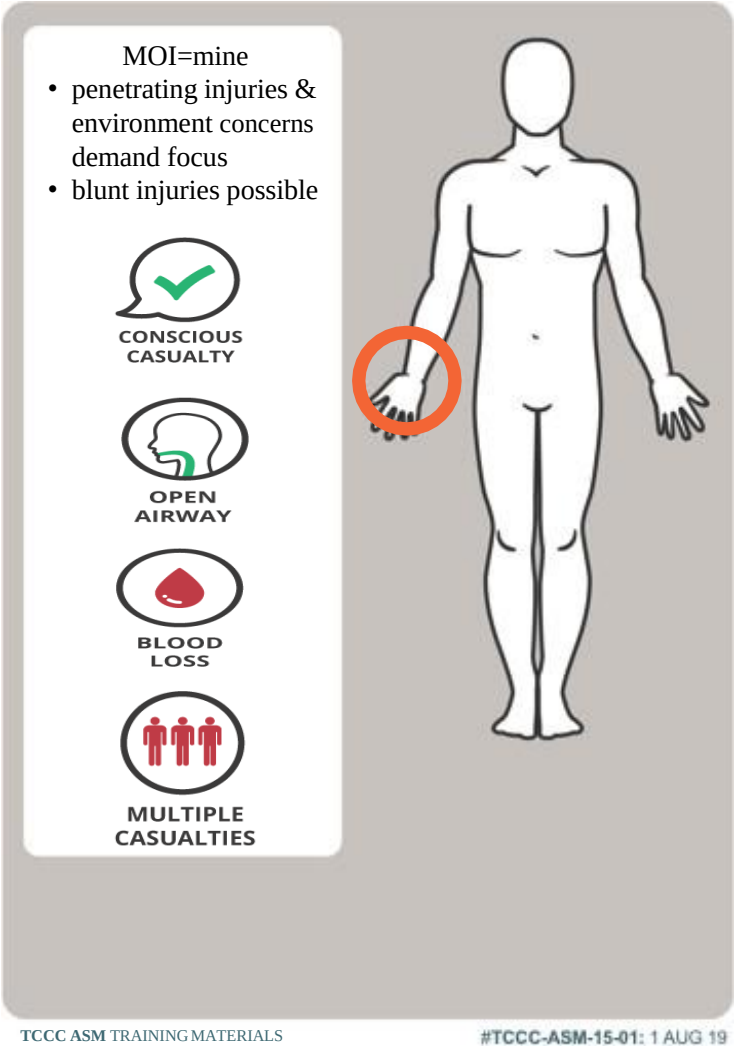
TCCC in the Maritime Scenario 2

Phase 2: Tactical Field Care (TFC)-Exam

Patient 2

What you see/feel:

- Alert, good eye contact, is in water above the knee
- Pulsating bleeding from his wrist
- Responsive, speaks clearly without difficulty
- Cool when touching skin



TCCC in the Maritime Scenario 1

Phase 2: Tactical Field Care (TFC)

Patient 2 - TFC Critical Actions

- M:
  - Right Arm Tourniquet
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression, chest seal placement as needed
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

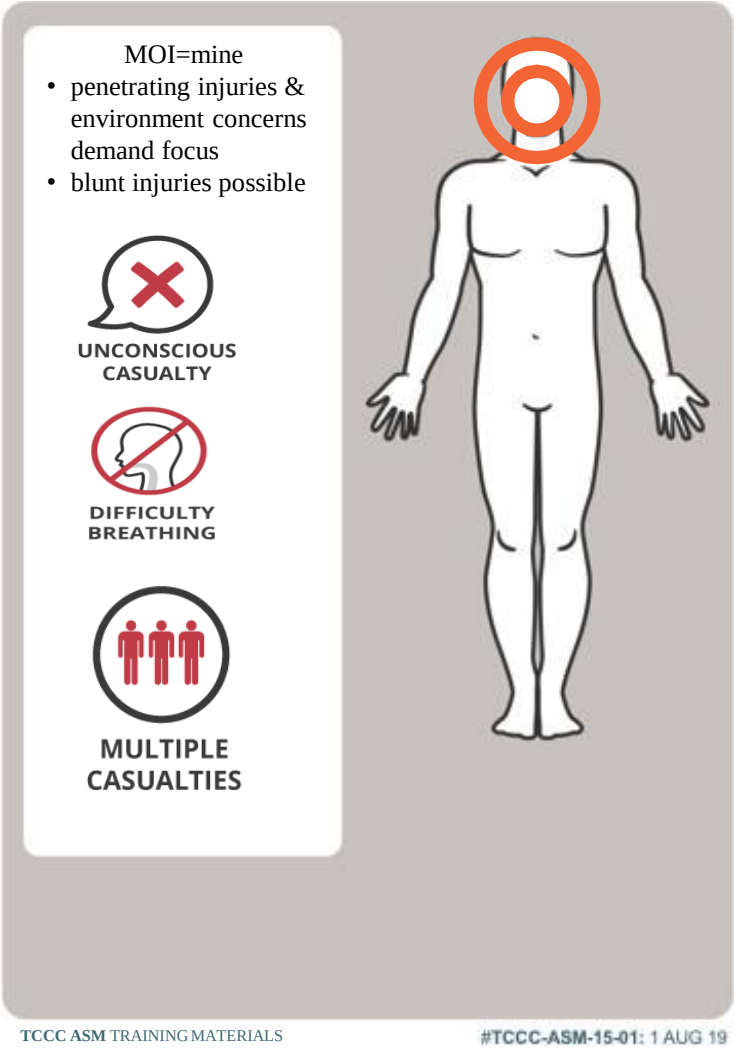
TCCC in the Maritime Scenario 2

Phase 2: Tactical Field Care (TFC)-Exam

- **Patient 3**

What you see/feel:

- Unconscious, lying face down in water
- Flip over-not breathing, no obvious bleeding
  - Perform airway maneuvers-starts breathing, lay in position to examine
- Responsive to touch, skin feels cool
- Pulse 100 bpm



TCCC in the Maritime Scenario 1

Phase 2: Tactical Field Care (TFC)

Patient 3 - TFC Critical Actions

- M:
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression, chest seal placement as needed
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

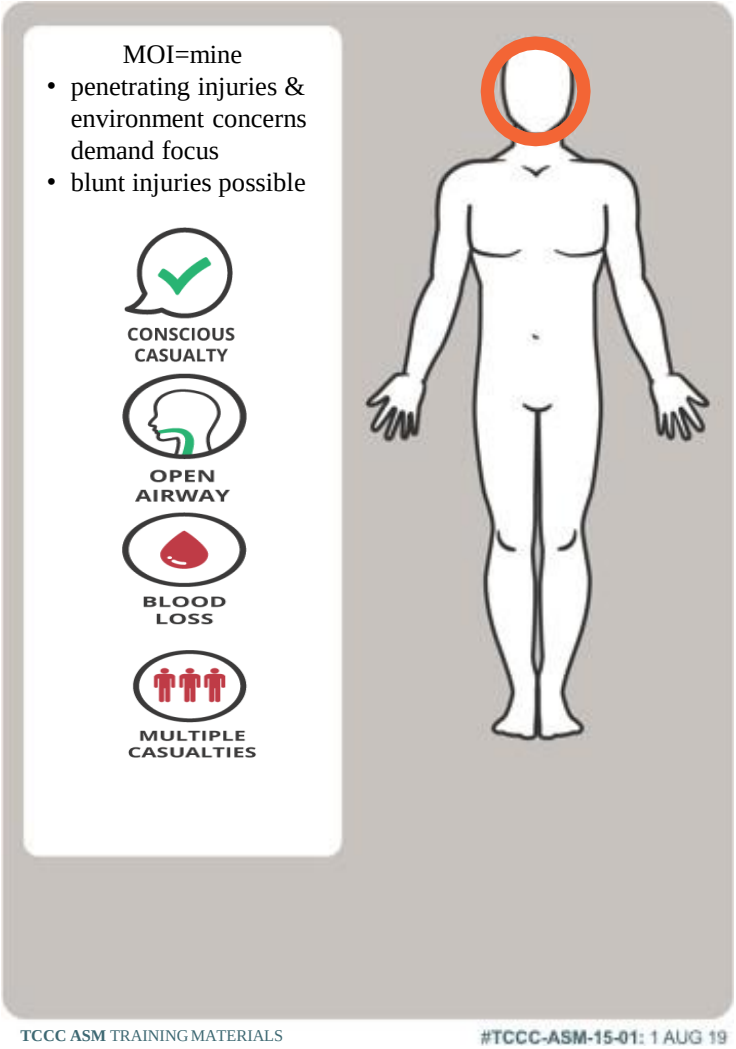
TCCC in the Maritime Scenario 2

Phase 2: Tactical Field Care (TFC)-Exam

- **Patient 4**

What you see/feel:

- Conscious, bleeding from the face
- Superficial headwound, no deformities
- Assess mental status
- Hypothermia prevention



TCCC in the Maritime Scenario 1

Phase 2: Tactical Field Care (TFC)

Patient 4 - TFC Critical Actions

- M:
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression, chest seal placement as needed
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Dressing to facial wound
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

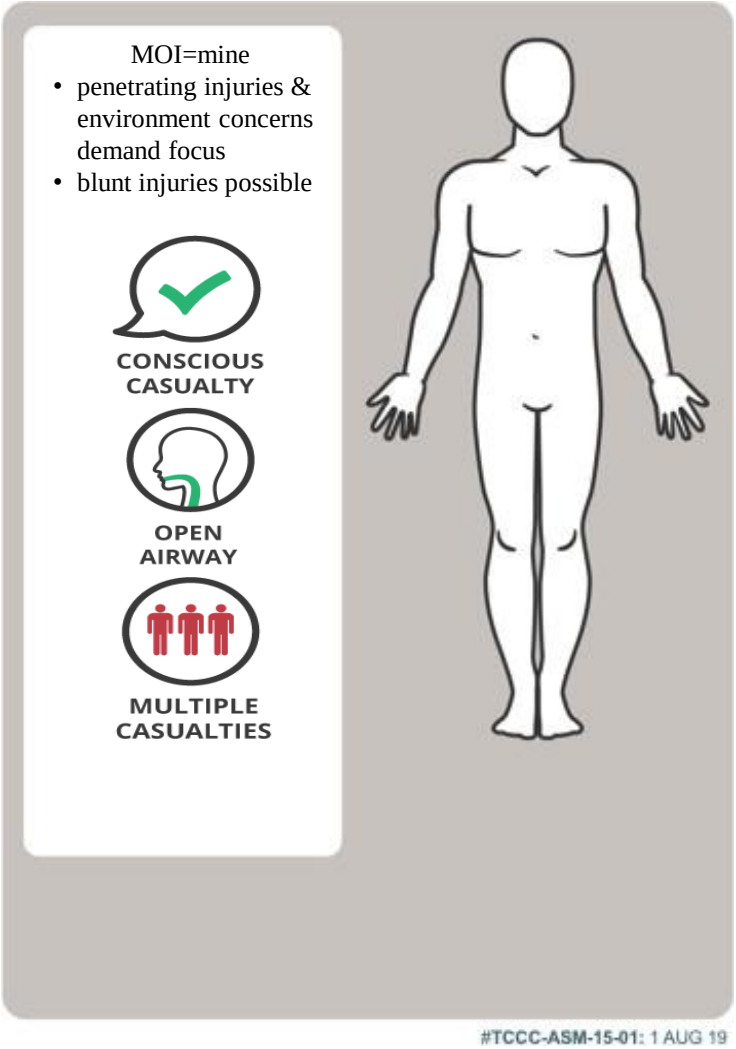
TCCC in the Maritime Scenario 2

Phase 2: Tactical Field Care (TFC) - Exam

Patient 1

What you see/feel:

- Conscious, talking, in water above the knees
- No visible bleeding or gross deformities noted
- Instruct patient to self-treat minor wounds
- Assist with Patient 4



TCCC in the Maritime Scenario 2

Phase 2: Tactical Field Care (TFC)

Patient 1- TFC Critical Actions

- M:
  - ❑ Check for other massive hemorrhage
- A:
  - ❑ Assess airway
  - ❑ Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - ❑ Assessment of breathing
  - ❑ Needle decompression, chest seal placement as needed
  - ❑ Oxygen administration
- C:
  - ❑ Recognize hemorrhagic shock
  - ❑ Intravenous/Intraosseous (IV/IO) access
  - ❑ Tranexamic Acid (TXA) administration*
  - ❑ Blood transfusion/activation of walking blood bank*
- H/H:
  - ❑ Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - ❑ Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - ❑ Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - ❑ Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - ❑ Inspect and dress wounds
  - ❑ Look for missed wounds
  - ❑ Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - ❑ TCCC documentation
  - ❑ Vital sign monitoring
  - ❑ Frequent reassessment

TCCC in the Maritime Scenario 2

Phase 2->3

Pt Movement / En Route Care

Pt Movement/En Route Care Critical Actions

- Patient packaging
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Cervical-spine precautions
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

Scenario 2b: Patient Injuries – 22 yo Female

- Unconscious
- Breathing: severe, progressive shortness of breath (rapid breathing)
- Lying in cold water, approximate depth of 10 cm

**Space has minimal flooding **

TCCC in the Maritime Scenario 2

Phase 1: Care Under Threat at Sea (CUT)

Patient - CUT Critical Actions

- Check scene safety
- Assess for Massive Hemorrhage & apply tourniquet
- Remove patient from compartment to safe space
- Address hypothermia

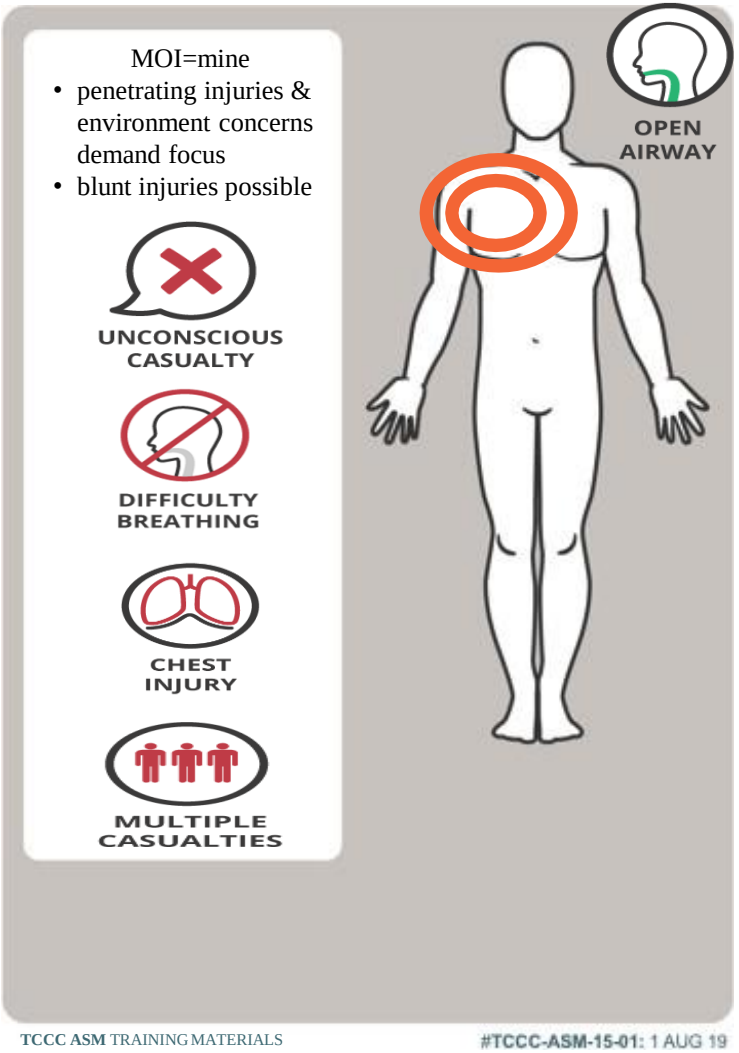
TCCC in the Maritime Scenario 2

Phase 2: Tactical Field Care (TFC)-Exam

Patient ??

What you see/feel:

- Patient moved to medical clearing area
- Unconscious, no obvious airway compromise but patient's breathing is very labored
- Right side of chest bruised and has poor rise, no visible penetrating injury
- Radial pulse is rapid, thread
- No response to stimuli, shivering with wet clothing
- No signs of fractures/gross deformities but multiple bruises noted to arms, legs
- VS: HR 133, BP 70/48, RR 32, Pulse Ox 87%, T 35°C (rectal)



TCCC in the Maritime Scenario 2

Phase 2: Tactical Field Care (TFC)

Patient - TFC Critical Actions

- M:
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression-pt has Tension PTX w/o penetrating wound
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
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- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

TCCC in the Maritime Scenario 2

Phase 2->3

Patient Movement/En Route Care

Patient ??-Movement/ERC Critical Actions

- Patient packaging
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Cervical spine precautions
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

TCCC in the Maritime Scenario 2

End of Scenario 2

Scenario 3: Helicopter Crash
Case Background

- French LHD
- Battle stations
- Eastern Mediterranean
- Helicopter maneuvers

Image from Google.com search

TCCC in the Maritime Scenario 3

- A helicopter lands violently on the flight deck of a landing helicopter dock (LHD). It is an emergency landing without warning
- Two crewmembers from the helicopter are ejected (patient #1 and #2)
- The other members of the helicopter crew show no injuries
- One person working on the flight deck was sprayed with flaming fuel (patient #3)

Response Considerations

- Can you access casualties right away?
- When can you access the flight deck?
- Consideration *before accessing flight deck*:
  - How many stretcher bearers do you have?
  - How do you move patients?
  - Where is the nearest first aid station located?
  - Supplies available and undamaged? What does the team carry?
  - Day or night conditions (light)?
  - What is size & composition of medical response team? Level of training?
  - Can you get confirmation of # of casualties?
  - Protect yourself (personal safety gear), confirm your position and wait for access clearance
  - What is the pre-planned patient movement route? Alternate "safe" routes according to damage control team?
  - En route care: who handles?

TCCC in the Maritime: Scenario 3

Phase 1: Care under Threat at Sea (CUT)

- You are the non-medical person providing medical care. You are cleared to access flight deck. You take action and tend to patients with major/life-threatening injuries.
- You've identified the following three casualties:
  - **Patient 1**: He is a co-pilot. He was ejected from the Helo. He is conscious and screaming from pain. He has a right leg fracture with pulsating bleeding.
  - **Patient 2:** He is pilot of the helicopter. He fell from 2 meters (6.5 ft.) trying to escape from the helicopter. He has an acute pain from his back and cannot move his legs.
  - **Patient 3**: He's a flight deck operator. He's conscious, burned (9%) front side of the thorax and 1% circular burn of the neck.

TCCC in the Maritime: Scenario 3

Phase 1: Care under Threat at Sea (CUT)

- At the point of injury – what is the next step?
- Who gets treated first?
  - The casualty with hemorrhage is treated first
  - He is the most important to treat immediately – he could bleed to death quickly
- Where should you treat the patient?
  - Location is now safe, you may treat within the safe zone on the flight deck

TCCC in the Maritime Scenario 3

Phase 1: Care Under Threat at Sea (CUT)

Patient 1 - CUT Critical Actions

- Check scene safety
- Assess for Massive Hemorrhage & apply tourniquet
- Remove patient from compartment to safe space
- Address hypothermia

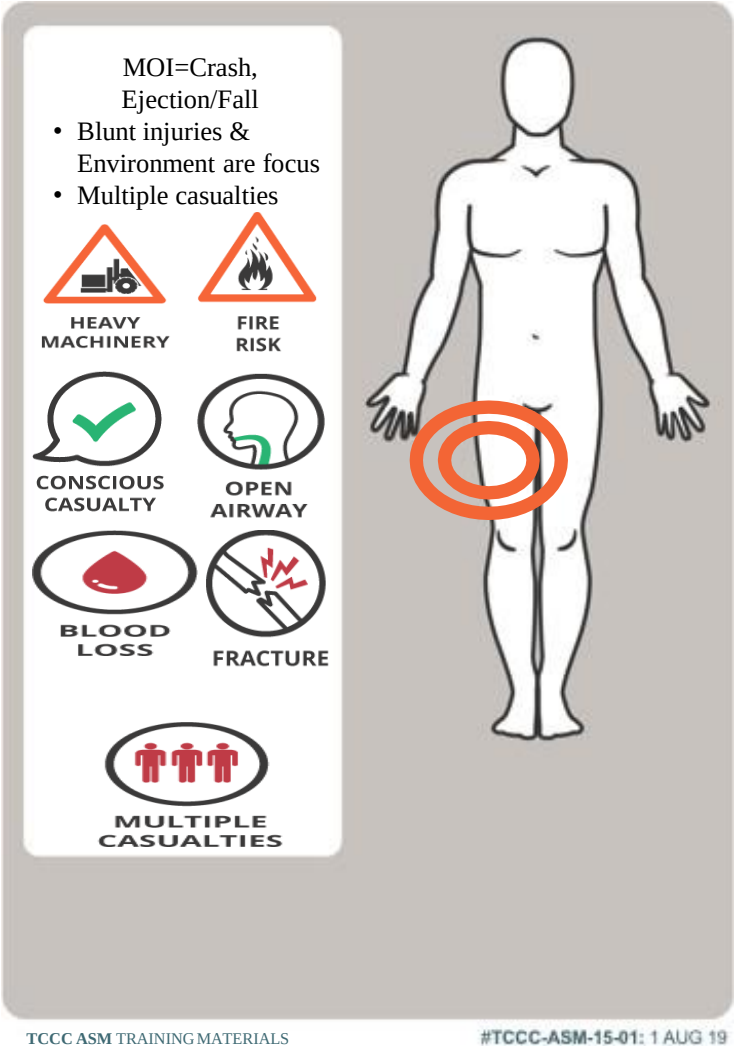
TCCC in the Maritime Scenario 3

Phase 2: Tactical Field Care (TFC)-Exam

Patient 1

What You See/Feel:

- Alert, screaming, pulsatile bleeding from deformed RT thigh w/visible bone
- Patient is breathing OK
- Wrist pulse strong, rapid



TCCC in the Maritime Scenario 3

Phase 2: Tactical Field Care (TFC)

Patient 1 - TFC Critical Actions

- M:
  - Right Thigh Tourniquet
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression, chest seal placement as needed
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

PT 1: Movement/ERC Care Critical Actions

- Patient packaging
  - Check immobilization (broken leg)
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Cervical-spine precautions
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

TCCC in the Maritime Scenario 3

Phase 1: Care Under Threat at Sea (CUT)

Patient 2 - CUT Critical Actions

- Check scene safety
- Assess for massive hemorrhage
- Remove patient from compartment to safe space
- Address hypothermia

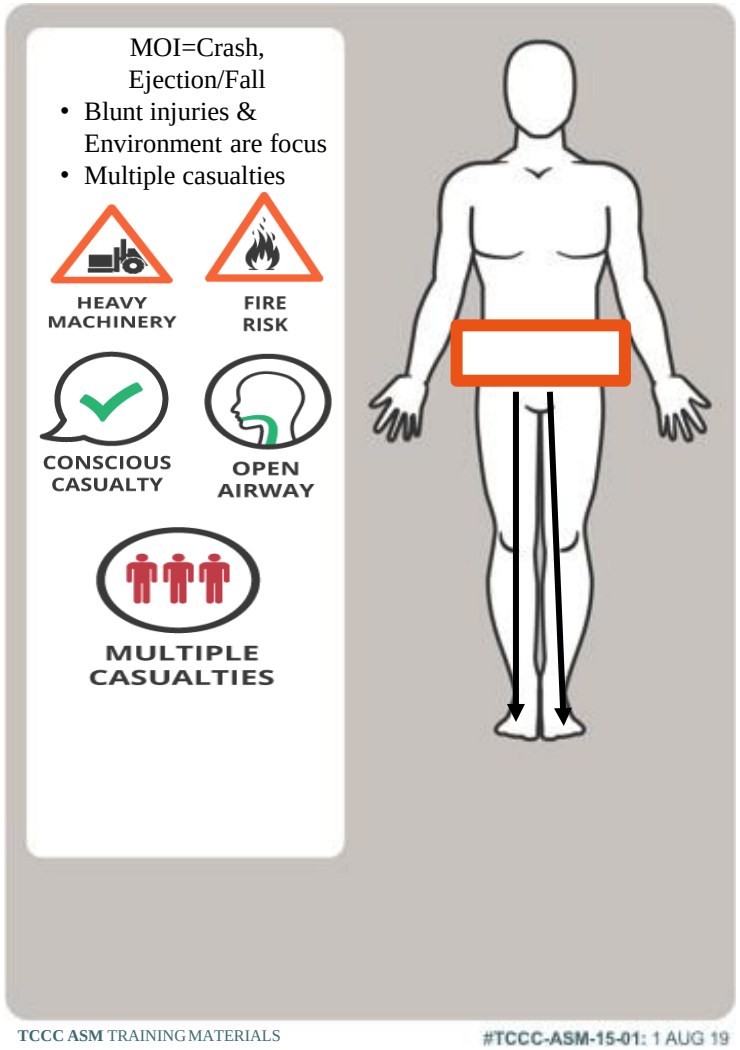
TCCC in the Maritime Scenario 3

Phase 2: Tactical Field Care (TFC)-Exam

Patient 2

What You See/Feel:

- Alert, talking and cooperative, concerned about legs and back pain
- No visible bleeding
- Breathing unlabored
- Strong peripheral pulses
- He has an acute pain from his back and cannot move his legs



TCCC in the Maritime Scenario 3

Phase 2: Tactical Field Care (TFC)

Patient 2 - TFC Critical Actions

- M:
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression, chest seal placement as needed
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
  - Neurovascular exam
  - Spinal precautions
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

Patient 2-Movement/ERC Care Critical Actions

- Patient packaging
  - Cervical-spine precautions
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

Patient 3 - CUT Critical Actions

- ❑ Check scene safety
- ❑ Assess for Massive Hemorrhage
- ❑ Remove patient from compartment to safe space
- ❑ Address hypothermia
  - Then, cover the burn area with dry, sterile dressings
  - Burn patients are at increased risk of hypothermia

TCCC in the Maritime Scenario 3

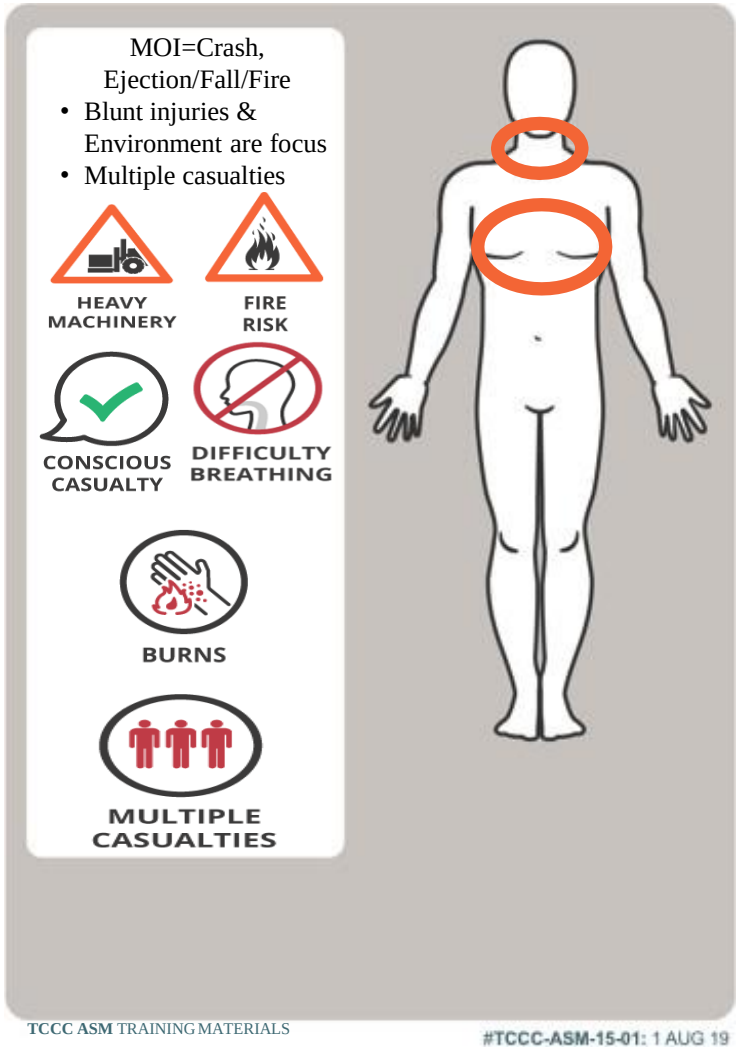
Phase 2: Tactical Field Care (TFC)-Exam

Patient 3

What You See/Feel:

- Alert, anxious appearing, visible burns/redness to neck and chest
- Audible wheezing
- Breathing fast and with labored effort
- Peripheral pulses strong bilaterally
- Burned (9%) front side of the thorax and 1% circular burn of the neck

***Recommend use of Burn shield (if available) - used to stop the burning process. However, burn shield should not be used more than 20 min, as it can increase risk of hypothermia. Recommend use of burn shield prior to application of sterile dressings.*



TCCC in the Maritime Scenario 3

Phase 2: Tactical Field Care (TFC)

Patient 3 - TFC Critical Actions

- M:
  - Check for massive bleeding (none)
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Obstructive edema: intubation and oxygen, *cricothyrotomy (aka coniotomy or emergency airway) if not possible*
  - Needle decompression, chest seal placement as needed
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
  - *Burn patients are at increased risk of hypothermia. Cover the burn area with dry, sterile dressings*
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - *Use "burnshield ®"(or other emergency burn care blanket) only on burned surface (maximum 20 min)*
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

Patient 3 – Movement/ERC Critical Actions

- Patient packaging
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Cervical-spine precautions
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

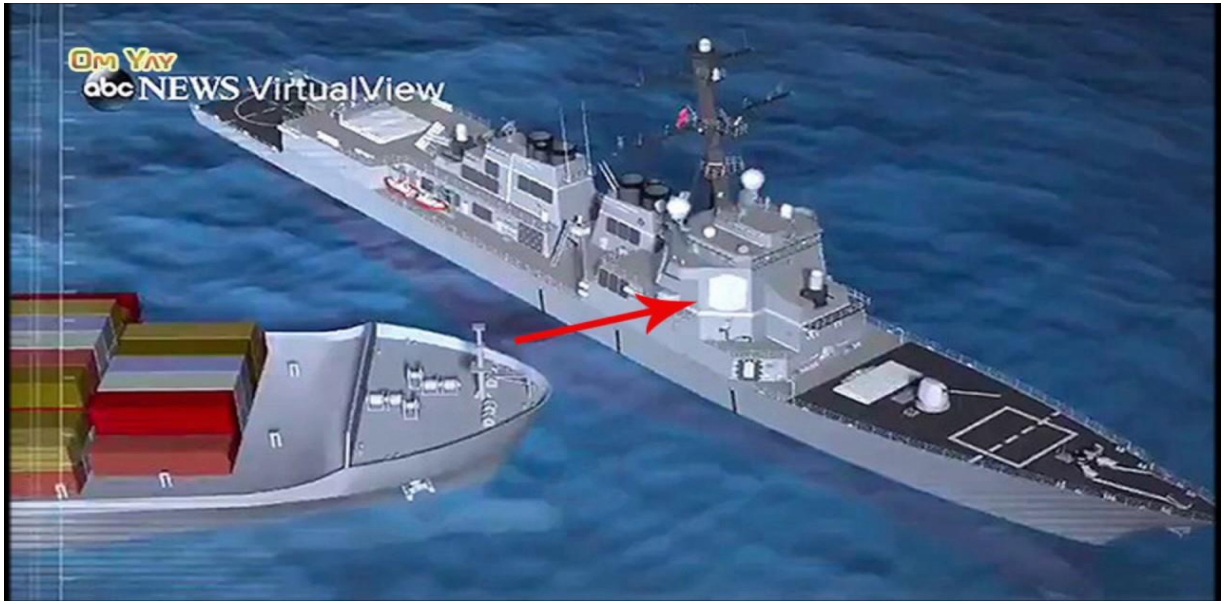
***Patient 3 can walk with help ***

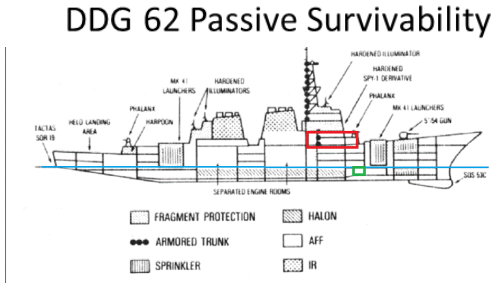
TCCC in the Maritime: Scenario 3

End of Scenario 3

Scenario 4: Collision
Case Background

- **Scenario is based off the USS FITZGERALD (DDG-62) with MV ACX Crystal collisions in 2017**
- **Nighttime collision to starboard side**
- **7 U.S. Navy sailors were killed and 3 injured**





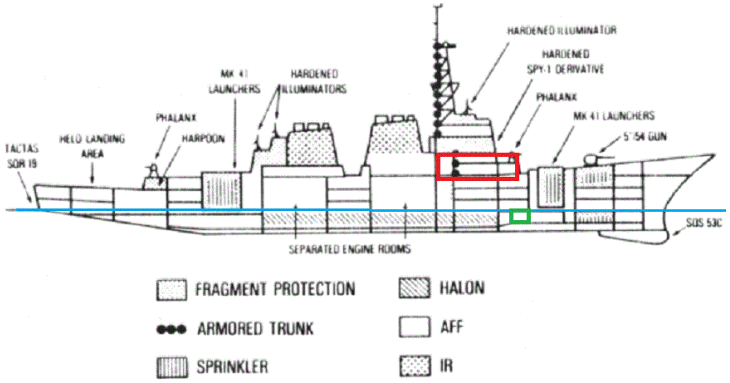
Images from Google.com search

TCCC in the Maritime Scenario 4

- **USS FITZGERALD is conducting routine low visibility maneuvering in the vicinity of the Gulf of Finland during BALTOPS**
- **DDG-62 is struck on starboard side by UK ship MV Hartland Point**
- **Large gash, starboard side, near the keel in the hull below the waterline-flooding occurred in crew berthing, radio room and a machinery space**
- **Two sailors injured in involved compartment**

Images from Google.com search

DDG-62 Passive Survivability



Response Considerations

- What types of injuries anticipated?
- What is size & composition of medical response team? Level of training?
- What resources and supplies available at nearest Battle Dressing Station? What does the team carry?
- What is the pre-planned patient movement route? Alternate "safe" routes according to damage control team?
- Status of blood supply? Walking blood bank?
- Proximity to next echelon of care? Patient movement assets?
- Who handles the En route care?

TCCC in the Maritime Scenario 4

Phase 1: Care Under Threat at Sea (CUT)

Patient 1 Injuries

22yo Male

- Right anterior chest with large area of bruising/ecchymosis
- Diffuse abdominal pain, some bruising noted
- Left leg deformity at ankle
- Multiple superficial cuts to both upper arms

Patient 2 Injuries

25yo Male

- Scattered bruising to all extremities and upper back
- Small Left facial bruise
- No obvious bleeding, patient is conscious but confused
- Found in water depth up to patients' waist (umbilicus)

**Significant flooding in compartment. Ambient water temp 11°C/52°F. Time from injury to assessment was 1 hour **

https://seatemperature.info/june/baltic-sea-water-temperature.html

TCCC in the Maritime Scenario 4

Phase 1: Care Under Threat at Sea (CUT)

Patient 1-CUT Critical Actions

- Check scene safety
- Assess for Massive Hemorrhage
- Remove patient from water-filled compartment
- Start addressing hypothermia when practical

Patient 2-CUT Critical Actions

- Check scene safety
- Asses for Massive Hemorrhage
- Remove patient from water-filled compartment
- Start addressing hypothermia when practical

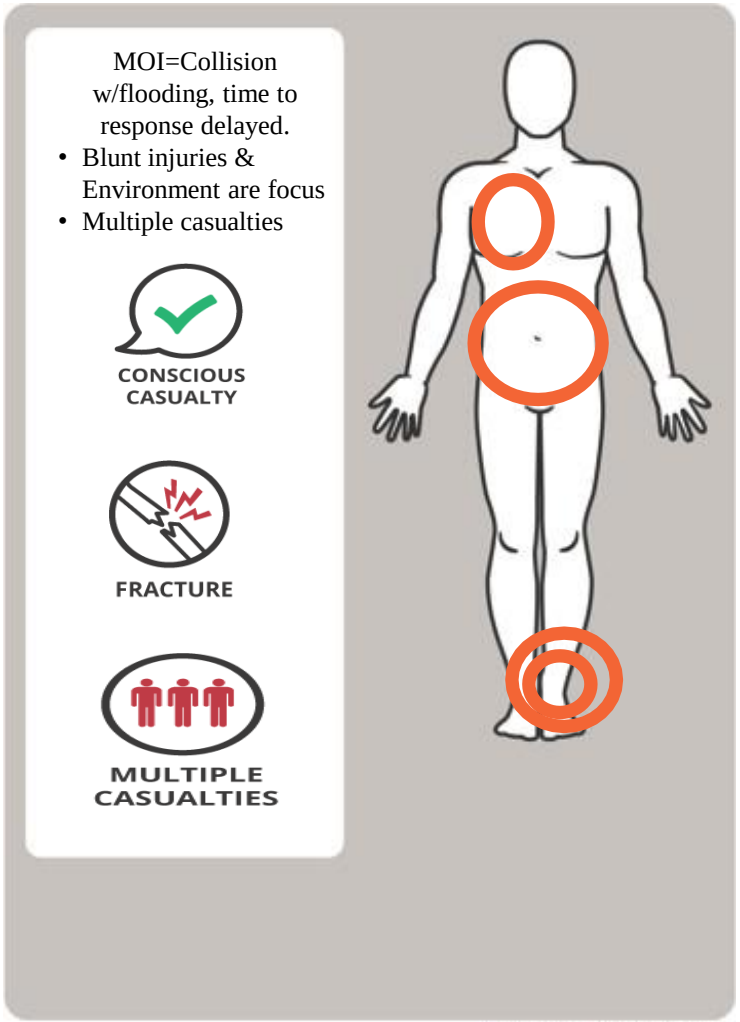
TCCC in the Maritime Scenario 4

Phase 2: Tactical Field Care (TFC)-Exam

Patient 1

What You See/Feel:

- Alert, interactive, shivering
- No obvious bleeding, LT ankle deformity is obvious
- Decreased breath sounds right chest with decrease chest wall rise compared to left
- Pulses are weak, thread, rapid in all extremities, elevated heart rate
- Left ankle with excessive outward rotation, swelling, normal contour lost, ankle pulse present but weak. Skin cool to touch throughout.



TCCC in the Maritime Scenario 4

Phase 2: Tactical Field Care (TFC)

Patient 1 - TFC Critical Actions

- M:
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression & securing of needle/device
  - Oxygen administration if able
- C:
  - Recognize hemorrhagic shock
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

TCCC in the Maritime Scenario 4

Phase 2->3

Patient Movement/En Route Care

Patient 1- Movement/ERC Critical Actions

- Patient packaging
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Cervical-spine precautions
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station (BDS) and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

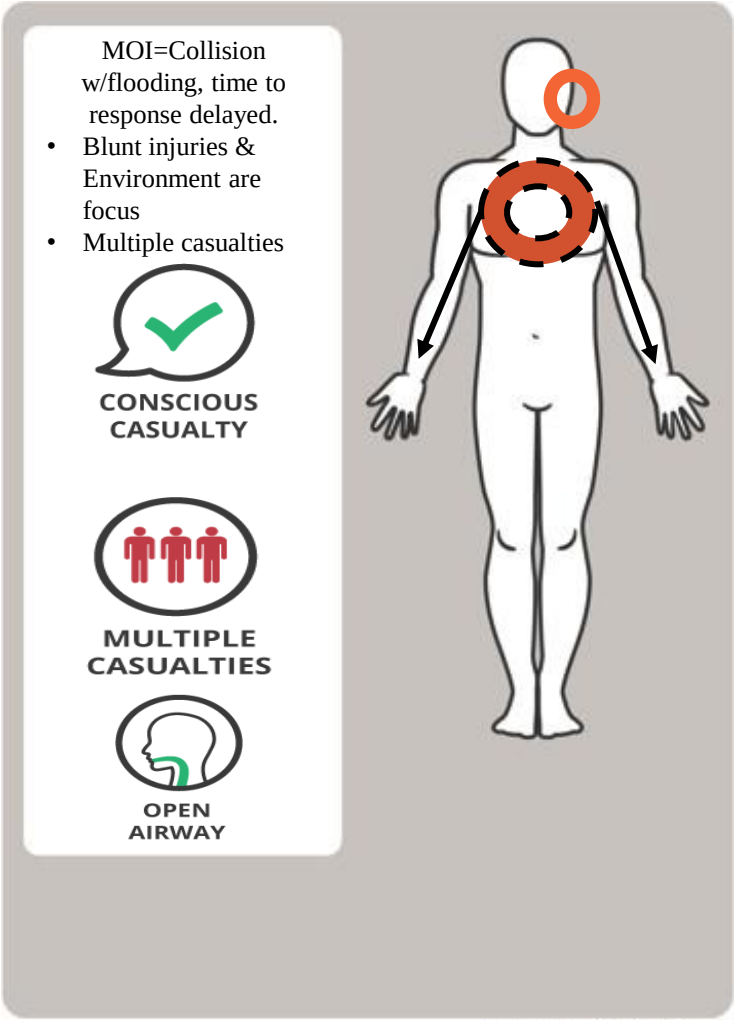
TCCC in the Maritime Scenario 4

Phase 2: Tactical Field Care (TFC)-Exam

Patient 2

What You See/Feel:

- Conscious, confused and agitated and shivering
- No bleeding seen
- Airway open, breathing is non-labored
- Palpable pulses in all extremities, slow heart rate (bradycardic)
- Skin very cool to touch



TCCC ASM TRAINING MATERIALS #TCCC-ASM-15-01: 1 AUG 19

TCCC in the Maritime Scenario 4

Phase 2: Tactical Field Care (TFC)

Patient 2 - TFC Critical Actions

- M:
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression, chest seal placement as needed
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Intravenous/Intraosseous (IV/IO) access
  - TXA (Tranexamic Acid) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU) –check for weapons or gear that could become a weapon
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment
  - TBI considerations

TCCC in the Maritime Scenario 4

Phase 2->3

Patient Movement/En Route Care

Patient 2- Movement/ERC Critical Actions

- Patient packaging
  - Properly deploy litter / TBI consideration, maintain head elevated greater than 30 degrees
  - Reassess all interventions after movement onto litter
  - Cervical-spine precautions
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention
  - Maintain adequate Bag Valve Mask ventilations during transport

TCCC in the Maritime Scenario 4

End of Scenario 4

Scenario 5: Boarding
Case Background

- **CTF (Combined Task Force) 150 Maritime Security Operations in the Arabian Gulf**
- **Mission is to search for illegal arm shipments in the Area of Operation (AO)**
- **Several NATO ships as part of a Task Group, to include an aircraft carrier with Role 2 level 4 Medical Treatment Facility**
- **A Vessel of Interest (VOI) is identified**
- **An Enhanced Boarding Party is dispatched to the vessel, with a 10 person team. All team members are TCCC trained**
- **Medical Pers on MOSHIP (Role 1 Level 2)**

Image source: https://ottawacitizen.com/news/national/defence-watch/royal-canadian-navys-new-maritime-tactical-operations-group-continues-to-grow

TCCC in the Maritime Scenario 5

- **Boarding commenced: 0500**
- **Access to ship by ladder; approx. 30 ft traverse from RHIB (Rigid hull inflatable boat) to main deck**
- **Boarding Party (BP) members board ship successfully**

Image source: https://ottawacitizen.com/news/national/defence-watch/royal-canadian-navys-new-maritime-tactical-operations-group-continues-to-grow

TCCC in the Maritime Scenario 5

- **First two members of BP advance along upper deck towards bridge**
- **Enter ship via starboard hatch**
- **Encounter enemy contact inside ship**
- **One member injured by gunfire and then collapses on deck**
- **Additional two members engage enemy**
- **Continued enemy interaction for next 10 mins with eventual elimination of immediate threat**

Image source: https://ottawacitizen.com/news/national/defence-watch/royal-canadian-navys-new-maritime-tactical-operations-group-continues-to-grow

Response Considerations

- Situational awareness/threats?
- What types of injuries anticipated?
- Mission status/safety?
- What is size & composition of medical response team? Level of training?
- What resources and supplies available ? What does the team carry?
- What is the pre-planned patient movement route? Alternate COAs?
- Proximity to next echelon of care? Patient movement assets?
- Who handles the En route care?
- Status of blood supply? Walking blood bank?
- Mission status?

TCCC in the Maritime Scenario 5

Phase 1: Care under Threat at Sea (CUT)

Patient 1 Injuries – 24 yo Male

- Right shoulder/upper chest GSW with mild bleeding
- Left thigh GSW with moderate bleeding
- Otherwise alert and communicating

Limited space inside ship/enemy interaction/noise

Patient 1 - CUT Critical Actions

- Check scene safety
- Assess for Massive Hemorrhage (tourniquet)
- Remove patient from compartment to safe space/area
- Address hypothermia
- Remove weapons/coms device
- Maintain tactical situational awareness/safety

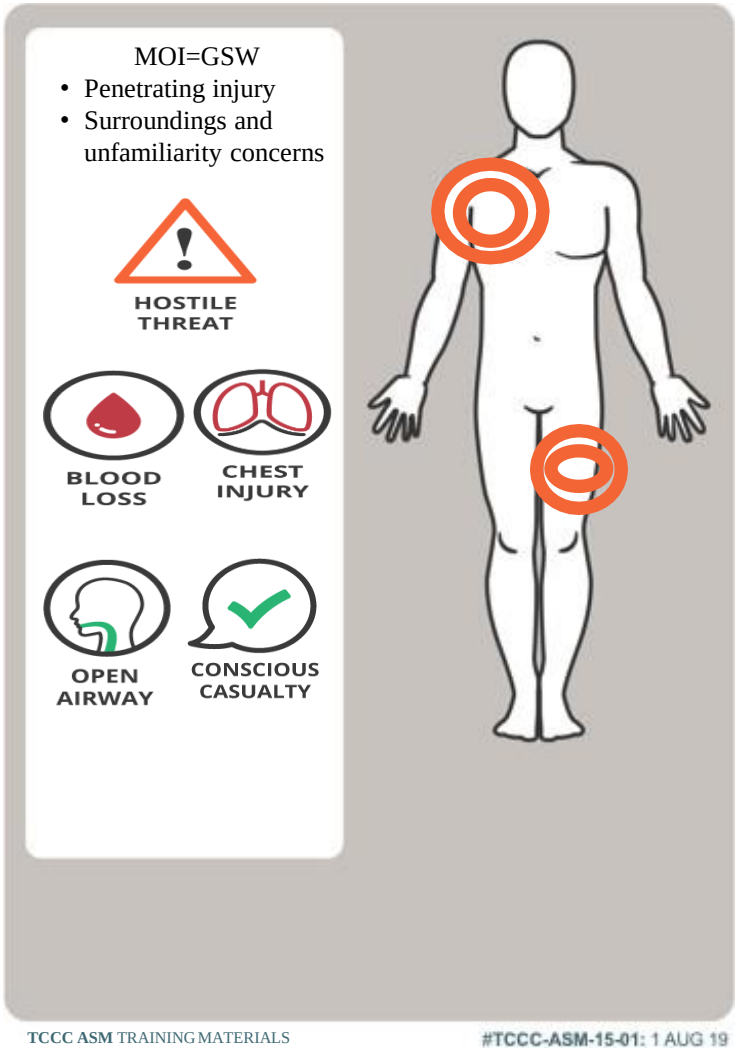
TCCC in the Maritime Scenario 5

Phase 2: Tactical Field Care (TFC)-Exam

Patient 1

What You See/Feel:

- Alert, communicates clearly
- Bleeding from chest and leg injuries- significant amount from leg.
- B: Chest asymmetry noted on right with increased RR
- Thready pulse (130+ bpm)
- D: Responsive



TCCC in the Maritime Scenario 5

Phase 2: Tactical Field Care (TFC)

Patient 1 - TFC Critical Actions

- M:
  - Left Thigh Tourniquet
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression, chest seal placement-anterior & posterior chest wounds
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
    - Are TQT's effective?
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

TCCC in the Maritime Scenario 5

Phase 2->3

Patient Movement/En Route Care

Patient 1 – Movement/ERC Critical Actions

- Patient packaging
  - Cervical-spine precautions
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with MOSHIP
  - Request additional support (2nd RHIB and medical support)
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

Scenario 5 *continued*: Boarding

Case Background

- **CTF150 Maritime Security Operations in the Arabian Gulf. Mission is to search for illegal arm shipments in the AO. Several NATO ships as part of a Task Group, to include an aircraft carrier with Role 2 level 4 Medical Treatment Facility**
- **A Vessel of Interest (VOI) is identified; an Enhanced Boarding Party is dispatched to the vessel, with a 10 person team. All team members are TCCC trained**
- **Medical Pers on MOSHIP (Role 1 Level 2)**

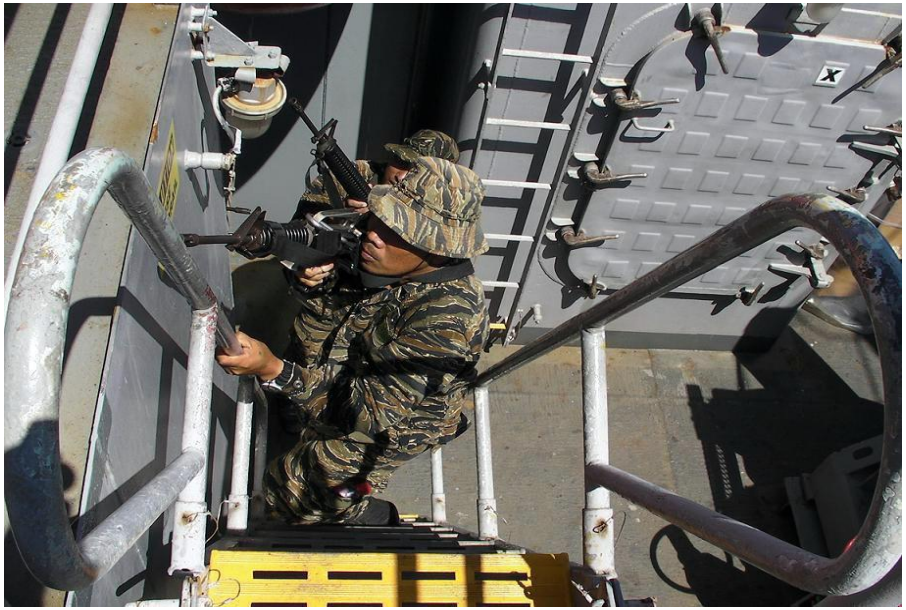
Image source: https://ottawacitizen.com/news/national/defence-watch/royal-canadian-navys-new-maritime-tactical-operations-group-continues-to-grow

TCCC in the Maritime Scenario 5
Continued

- **Boarding commenced 0500**
- **Access to ship by ladder; approx. 30 ft traverse from RHIB to main deck**
- **Five Boarding Party members alight ship successfully**
- **6th member to climb ladder lost footing and fell 10 ft into RHIB sustaining injury**

Image source: https://ottawacitizen.com/news/national/defence-watch/royal-canadian-navys-new-maritime-tactical-operations-group-continues-to-grow

TCCC in the Maritime Scenario 5
Continued



- **Boarding commenced 0500**
- **Access to ship by ladder; approx. 30 ft traverse from RHIB to main deck**
- **All 6 Boarding Party members alight ship successfully**
- **Kinetic engagement of VOI crew. Boarding Party advance forward and up towards bridge.**
- **Team member to climb lost footing climbing up ladder and fell 15 ft into RHIB sustaining injury**

Image source: https://ottawacitizen.com/news/national/defence-watch/royal-canadian-navys-new-maritime-tactical-operations-group-continues-to-grow

Response Considerations

- Situational awareness?
- Mission status/safety?
- What types of injuries anticipated?
- What is size & composition of medical response team? Level of training?
- What resources and supplies available? What does the team carry?
- What is the pre-planned patient movement route? Alternate COAs ?
- Proximity to next echelon of care? Patient movement assets?
- Who handles the En route care?

TCCC in the Maritime Scenario 5

Phase 1: Care under Threat at Sea (CUT)

- **<u>Patient 2 Injuries – 28 yo Male</u>**

- Visible bruising to right side of head
- Severe bruising right lateral/anterior chest
- Contusion right flank
- Right leg externally rotated and shortened

Sea-state 2/RHIB stable/No immediate threat from ship

Patient 2 - CUT Critical Actions

- Check scene safety
- Assess for Massive Hemorrhage
- Relocate patient to “treatment area” of RHIB
- Remove weapons/coms device
- Address hypothermia
- Maintain tactical situational awareness

Sea-State: 2/No enemy from top-side

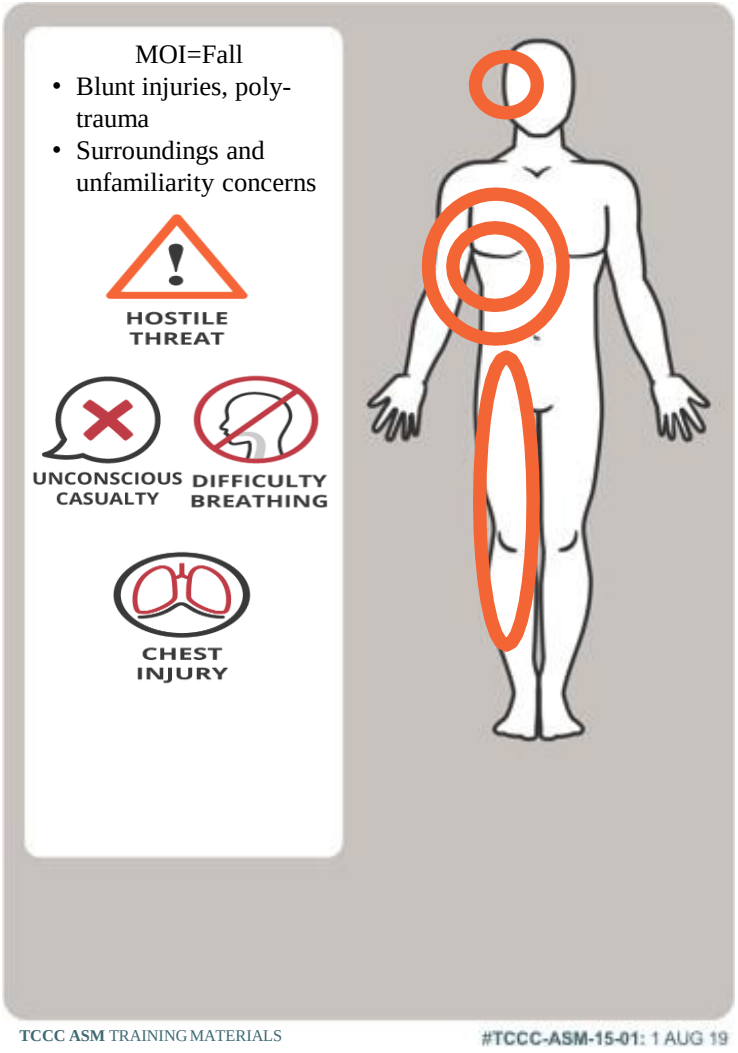
TCCC in the Maritime Scenario 5

Phase 2: Tactical Field Care (TFC)-Exam

Patient 2

What You See/Feel:

- Unconscious, obvious RT leg injury. No bleeding seen.
- Airway semi-open, secretions from mouth, high pitched wheezing
- Reduced rise of chest-wall, increased Work Of Breathing, increased RR
- Weak, thready pulse (140's bpm)
- Skin cool and mottled
- Right leg externally rotated and shortened



TCCC in the Maritime Scenario 5

Phase 2: Tactical Field Care (TFC)

Patient 2 - TFC Critical Actions

- M:
  - Check for other massive hemorrhage
    - If concern noted due to MOI, place pelvic binder
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing
  - Needle decompression & securing of device
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
    - Placement of pelvic binder
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU): C-Collar placement
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

TCCC in the Maritime Scenario 5

Phase 2->3

Patient Movement/En Route Care

Patient 2 – Movement/ERC Critical Actions

- Patient packaging
  - Cervical-spine precautions
  - Head elevated for TBI concerns
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with MOSHIP
  - Request additional support (2nd RHIB and Med Tech)
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

End of Scenario 5

Scenario 6: Pick Up from Water
Case Background

- **Scenario is based off the German Patrol Boat P 8732 KRONSHAGEN performing a NATO mission in cold Northern waters**
- **Another vessel within the mission, P 8729 MELSDORF had an explosion on board caused by a missile strike and sent distress messages**
- **P 8729 MELSDORF listed and sank**

TCCC in the Maritime Scenario 6

- All available NATO units in vicinity started to search for survivors
- P 8732 KRONSHAGEN comes close to a sailor floating in a lifejacket
- The crewmates retrieve the victim to provide medical care
- Casualty is wearing BDU and life jacket
  - No gloves, cap, nor dry suit
  - He is exhausted
  - Skin is pale, except for both of his hand
  - Neck and chin - skin appears red with single blisters

Response Considerations

- What types of injuries anticipated?
- What is size & composition of medical response team? Level of training?
- What resources and supplies available at nearest Battle Dressing Station? What does the team carry?
- What is the pre-planned patient movement route? Alternate "safe" routes according to damage control team?
- Status of blood supply? Walking blood bank?
- Proximity to next echelon of care? Patient movement assets?
- Who handles the En route care?

TCCC in the Maritime Scenario 6

Phase 1: Care under Threat at Sea (CUT)

Patient Injuries – 25 yo Male

- Immersion & Submersion: near drowning and hypothermia
- Chemical burns
- Dehydration
- Conscious
- Open airway, difficulty breathing

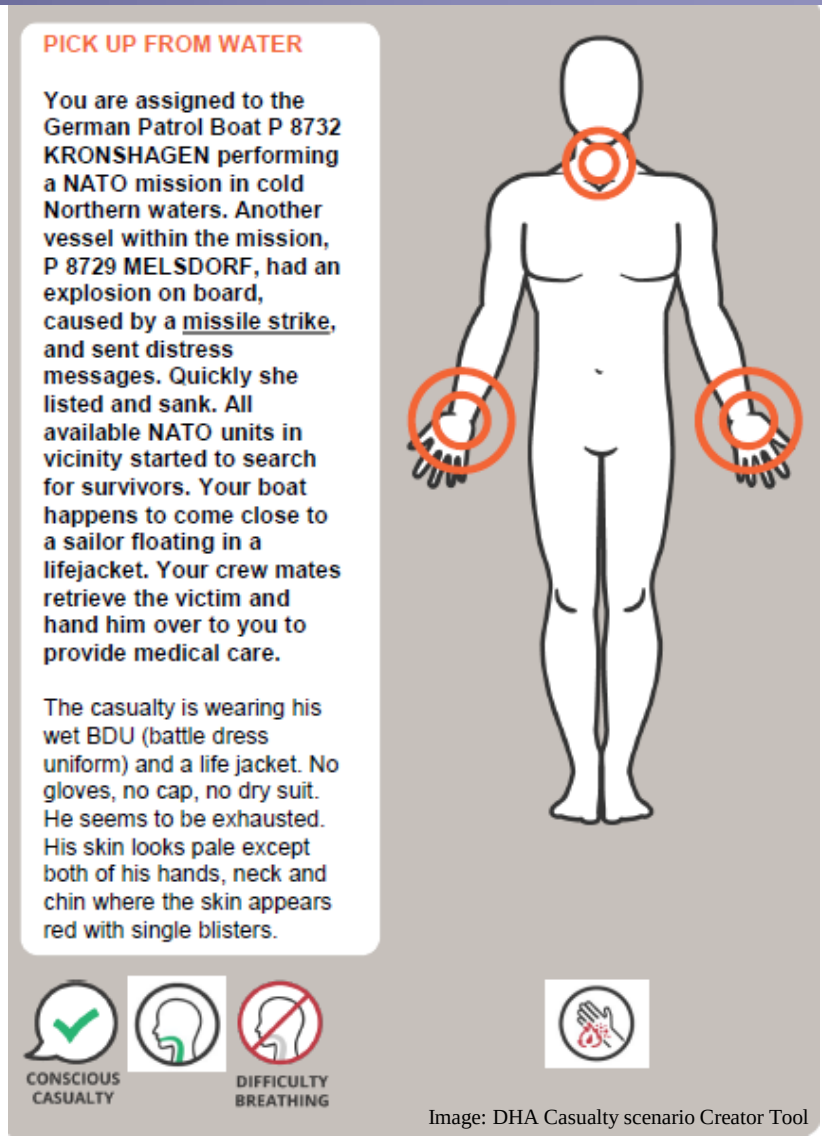


Image: DHA Casualty scenario Creator Tool

Patient - CUT Critical Actions

- Check scene safety
- There is no imminent threat to you or your unit at that moment
- Assess for Massive Hemorrhage
- Remove patient from compartment to safe space
- Address hypothermia

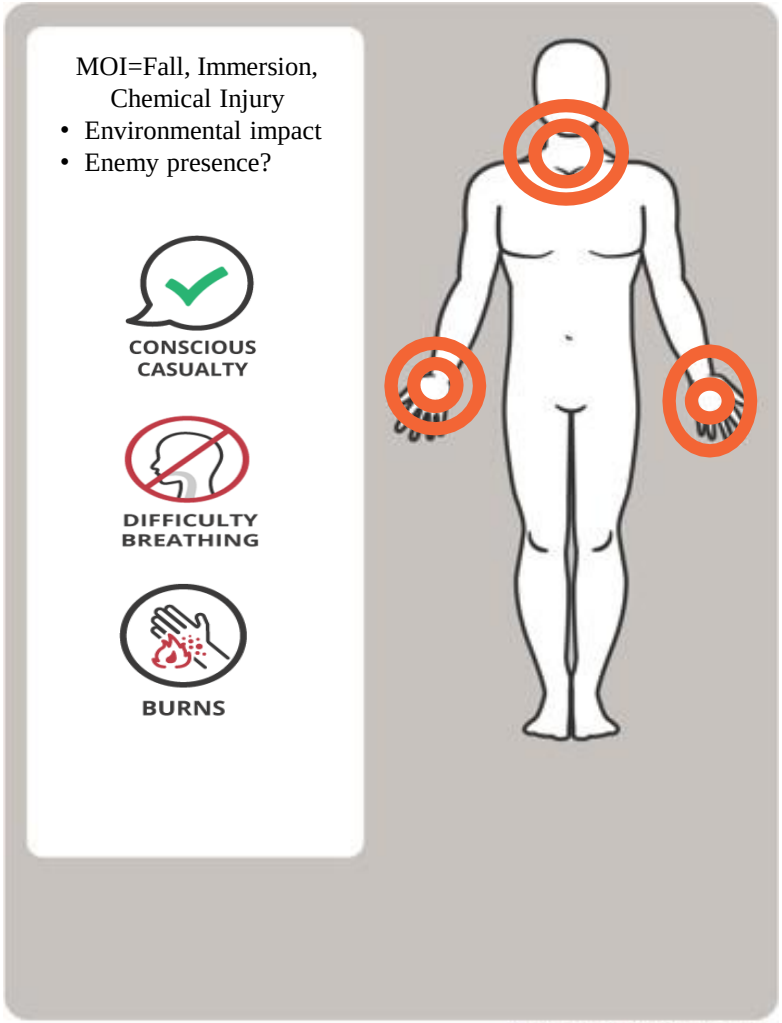
TCCC in the Maritime Scenario 6

Phase 2: Tactical Field Care (TFC)-Exam

Patient 2

What You See/Feel:

- Shivering, wet, conscious, fatigued w/erythema and blistering about face
- No visible bleeding
- Maintaining airway
- Tachypneic, tachycardic
- Palpable pulse, rate difficult to asses due to shivering
- BP/POX not obtainable/no reading
- No gross deformities



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TCCC in the Maritime Scenario 6

Phase 2: Tactical Field Care (TFC)

Patient - TFC Critical Actions

- M:
  - Check for other massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, or NasoPharyngeal Airway (NPA) as needed
  - Chemical burn considerations
- R:
  - Assessment of breathing
  - Needle decompression, chest seal placement as needed
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
  - Intravenous/Intraosseous (IV/IO) access
  - Tranexamic Acid (TXA) administration*
  - Blood transfusion/activation of walking blood bank*
- H/H:
  - Assessment of mental status Alert/Verbal/Pain/Unresponsive (AVPU)
  - Hypothermia prevention (Remove wet clothes, Mylar blanket)
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - Intramuscular/Intravenous (IM/IV) Ertapenem (No oral meds with altered mental status)
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
  - Splint fracture/dislocation and check pulses
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

TCCC in the Maritime Scenario 6

Phase 2->3

Patient Movement/En Route Care

Patient – Movement/ERC Critical Actions

- Patient packaging
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Cervical-spine precautions
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

End of Scenario 6

Scenario 7: FFG WBIED Attack
Case Background

- **Scenario is based off the USS COLE, a guided missile destroyer (DDG-67), which was attacked by a small craft laden with explosives by suicide bombers on 12 October 2000**
- **Seventeen U.S. Navy sailors were killed and 39 injured**

Images from Google.com search

Case Background USS Cole

Table 1 Fatalities

Fatality	Potential Survival (AFIP)	Mechanism	Injuries	Treatment, if any
1	Unsurvivable	Blunt force	Brain contusions; skull fractures; intracranial hemorrhage; clavicle Fx; rib fractures; lacerated liver and right lung; compound Fxs of humerus, tibia, fibula	None, rapid death
2	Potentially survivable	Blunt force	Maxilla Fx; multiple rib Fxs; lacerations of liver, kidney, spleen; bilateral ankle Fxs	Abdominal exploration, splenectomy, left ankle fracture pin, died 8 h postinjury
3	Unsurvivable	Blunt force	Skull and facial Fxs, partial brain avulsion, multiple rib Fxs, lung contusions, pelvic Fx, contusions of small and large intestines, compound radial Fx, right femur Fx, bilateral fibula Fxs	None, rapid death
4	Potentially survivable	Blunt force	Depressed skull Fx and basal skull Fx with brain contusions and subarachnoid hemorrhage, perforated tympanic membranes, Fxs of mandible and maxilla, lung contusions, left radius and femur Fx, bilateral fibula Fx, right ankle Fx	First aid only, died in minutes
5	Unsurvivable	Blunt force	Atlanto-occipital Fx; mandible Fx; multiple rib Fxs; left clavicle Fx; pelvic Fxs; T7–T8 Fx with spinal cord transection; lacerations of heart, lung, aorta, liver, spleen, kidneys, adrenals; laceration of diaphragm with intestinal hernia; right tibia and fibula Fxs	None, instantaneous death
6	Unsurvivable	Blunt force	Extensive skull and facial Fxs; avulsion of brain; thyroid and cricoid cartilage Fxs; C5 Fx with spinal cord transection; multiple rib fractures; T8–T9 Fx with spinal cord transection; multiple pelvic Fxs; lacerations of the spleen, bladder, and rectum; right femur Fx; right tibia/fibula Fx; left radius/ulna Fx; left tibia/fibula Fx	None, instantaneous death
7	Unsurvivable	Blunt force	Fx of skull, base, and calvarium; multiple rib Fxs; lacerations of heart, aorta, and vena cava; T2 Fx; lacerations of liver, spleen, bladder; avulsion of pelvis, scrotum, and buttocks; multiple pelvic Fxs, right radius/ulna Fxs; bilateral Fxs of femurs, tibias, fibulas, and feet	None, instantaneous death
8	Unsurvivable	Blunt force	Extensive facial and skull fractures with brain avulsion; C4 Fx; hyoid and thyroid cartilage Fxs; multiple bilateral rib Fxs; bilateral clavicle Fxs; ruptured diaphragm; lacerations of heart, lungs, liver, spleen, and left kidney; Fxs of pelvis, sacrum, and T8; bilateral Fxs of humerus, radius, femurs, and ankles; left ulna Fx	None, instantaneous death
9	Unsurvivable	Blunt force	Extensive skull and facial Fxs, avulsion of brain, extensive fragmentation and evisceration of chest and abdominal organs, amputation of all extremities	None, instantaneous death
10	Unsurvivable	Blunt force	Basal and frontal skull Fxs; multiple Fxs of ribs, vertebrae, and coccyx; right scapula, humerus, tibia Fxs; bilateral ulna and feet Fxs	None, rapid death
11	Unsurvivable	Blunt force	Extensive injuries with avulsion of face and scalp; extensive skull fractures with avulsion of brain and neck organs; extensive injuries of chest, abdomen, and pelvis with avulsion of organs; Fx dislocation of T8, T11, and L5; Fxs of all extremities	None, instantaneous death
12	Unsurvivable	Blunt force	Multiple comminuted Fxs of the face and skull; partial anterior decapitation; bilateral multiple rib fractures; right clavicle Fx; extensive Fxs of sacrum and pelvis; Fxs of right humerus, femur, tibia/fibula; Fxs of left humerus, radius/ulna, fibula and both feet	None, instantaneous death
13	Unsurvivable	Blunt force	Basilar skull and nose Fxs, lacerations of left chest wall, rib Fxs, lung contusions	None, rapid death
14	Unsurvivable	Drowning	No traumatic injuries	None
15	Unsurvivable	Drowning	Contusions of the back, foreign material aspiration	None
16	Unsurvivable	Drowning	No traumatic injuries	None
17	Unsurvivable	Blunt force	Multiple bilateral rib fractures, lacerations of heart and lungs, lacerations of right diaphragm and liver, right tibia/fibula Fxs	None, rapid death

AFIP, Armed Forces Institute of Pathology; Fx, fracture.

Table 2 Survivor Injury Distribution[a]

Patient	Mechanism	Diagnoses	Treatment
1	Fragment injury	Upper extremity cellulitis and abscess	Debridement, antibiotics
2	Blunt blast injury	Closed mandible Fx, subconjunctival hemorrhage	Initial dental wiring, later nutrition support
3	Blunt blast injury, toxic gas exposure	Clavicle Fx, chest wall contusion, hemothorax, scapular Fx, splenic hematoma, inhalation injury, spinal column Fx, open upper extremity wound	Pleuracentesis, oxygen supplementation, sling, aortography, cerebral angiography, antibiotics, debridement
4	Blunt blast injury, fragment wounds	Concussion, multiple cutaneous shrapnel wounds	Observation
5	Blunt blast injury, primary blast injury	Perforated TM, superficial conjunctival laceration, corneal abrasions, concussion, tinnitus, musculoskeletal pain	Myringoplasty, cranial CT scan, audiologic examination, ophthalmologic examination and treatment
6	Blunt blast injury	Mandible Fx	Initial observation and fixation of fracture; later removal of hardware, treatment of infection, and fixation
7	Traumatic stress	Bereavement	Counseling, grief therapy
8	Blunt blast injury, burns, fragment wounds	Facial and upper extremity second-degree burns, keratitis, open wound of lower extremity, foreign bodies in conjunctiva venous thrombosis, wound infection	CT scan of head, removal of foreign bodies and singed lashes from eye, superficial burn wound care, debridement of wounds, duplex ultrasound
9	Blunt blast injury, fragment wound	Open wound/Fx of the foot/ankle	Debridement, antibiotics, below-knee amputation
10	Blunt blast injury, toxic gas exposure, fragment wound	Inhalation injury, first-degree foot burn, fragment wound of lower extremity	Observation, superficial wound care
11	Blunt blast injury	Multiple orbital Fx, subdural hematoma, cranial nerve IV palsy, scalp wound	Wound irrigation and closure, eye examination, head CT scan, observation
12	Blunt blast injury	Gluteal contusion, shoulder contusion	Observation
13	Blunt blast injury, toxic gas exposure	Knee contusion, inhalation injury, knee effusion	Observation
14	Primary blast injury	Perforated TM	Observation
15	Blunt blast injury	Extremity contusions	Observation
16	Blunt blast injury, burn, primary blast injury	First-degree facial burn, keratitis, perforated TM, tooth Fx	Observation, eye examination and medications, removal of singed lashes
17	Blunt blast injury, primary blast injury	Scaphoid Fx, perforated TM, hip contusion	Closed fracture reduction
18	Burn, fragment wounds	Facial first-degree burns, corneal foreign bodies, corneal thermal injury	Superficial burn wound care, eye examination and medication
19	Blunt blast injury, fragment wounds	Facial wounds, upper extremity contusions	Irrigation of wounds
20	Blunt blast injury	Ankle sprain	Ankle support
21	Blunt blast injury, primary blast injury	Closed head injury, perforated TM	CT scan of head
22	Fragment injury	Scalp wound	Irrigation and wound closure
23	Blunt blast injury	Lumbar sprain	Observation
24	Blunt blast injury	Contusion of foot	Observation
25	Blunt blast injury, fragment wounds	Facial wounds, multiple abrasions, conjunctival hemorrhage	Irrigation of wounds, superficial wound care, eye examination
26	Blunt blast injury	Pneumothorax, scapula Fx, ligamentous knee injury, rib Fxs, knee effusion	Chest tube, knee reconstruction, MRI of knee
27	Traumatic stress	Psychiatric trauma	Counseling and therapy
28	Blunt blast injury	Concussion	CT scan of head
29	Blunt blast injury	Concussion, scalp wound	CT scan of head, irrigation and closure of wound
30	Blunt blast injury	Elbow and knee sprain, gluteal contusion	Observation
31	Blunt blast injury, fragment wounds	Facial wounds, lower extremity contusion, teeth Fxs	Initial closure of lacerations; later splint, observation, dental care
32	Blunt blast injury	Open femur Fx, ligamentous knee injury	Intramedullary rod, debridement, irrigation of wound, knee reconstruction
33	Blunt blast injury	Open femur Fx, radius Fx, lip laceration	Intramedullary rod, debridement, wound irrigation, repair of lip laceration
34	Blunt blast injury, primary blast injury of the lung, fragment wounds	Dislocation of knee, primary blast injury of lung, pulmonary hematocele, pulmonary contusions, facial wounds, perforated TM, rib Fxs	CT scan of thorax, arteriography, wound irrigation and closure, supplemental oxygen
35	Blunt blast injury	Open tibia Fx, fibula Fx, tibial plateau Fx, comminuted ankle Fx	Intramedullary rod in tibia, ORIF tibial plateau Fx fixation, ORIF ankle Fx, skin graft, soleus flap, MRI and multiple radiography

TM, tympanic membrane; Fx, fracture; CT, computed tomographic; MRI, magnetic resonance imaging; ORIF, operative reduction and internal fixation.

[a] Type I, primary; type II, fragment; type III, blunt; type IV, toxic gases and burns.

Source: CDR Thomas P. Davis, Distribution and Care of Shipboard Blast Injuries (USS Cole DDG-67), J Trauma. 2003;55:1022–1028

TCCC in the Maritime Scenario 7

- A NATO Frigate is conducting daytime freedom of navigation mission
- The frigate engaged by multiple small, unmarked vessels challenging the ship
- The frigate is struck by smaller vessel laden with IED, striking port side, aft of ship above water line. Helicopter deck is inoperable (note new spot of MEDEVAC hoisting)
- Two sailors injured in involved compartment

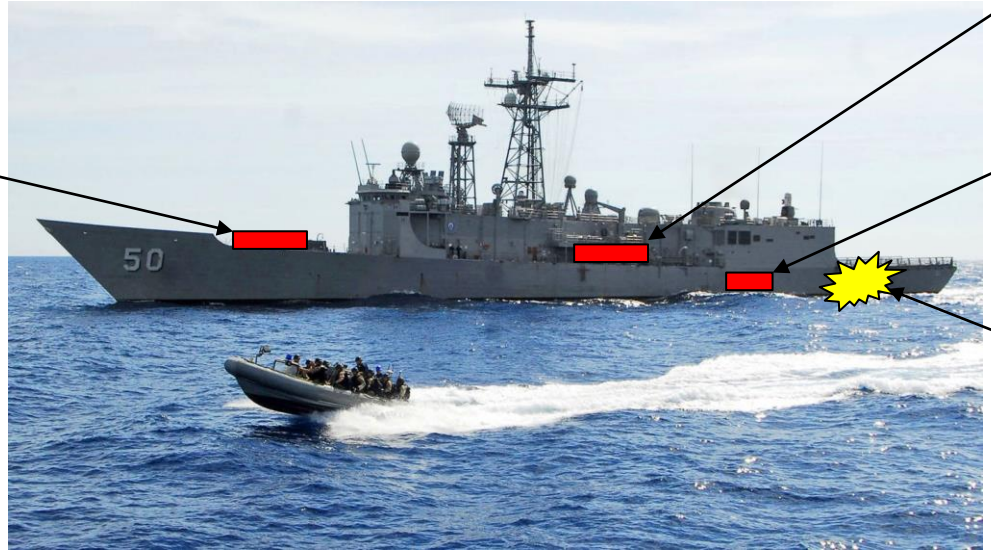


Image source: navalphotos.blogspot.com

Response Considerations

- What types of injuries anticipated?
- Where are the casualties located?
- What is size & composition of medical response team? Level of training?
- What resources and supplies available at nearest Battle Dressing Station? What does the team carry?
- What is the pre-planned patient movement route? Alternate "safe" routes according to damage control team?
- Status of blood supply? Walking blood bank?
- Proximity to next echelon of care? Patient movement assets?
- Who handles the En route care?

TCCC in the Maritime Scenario 7

Phase 1: Care Under Threat at Sea (CUT)

Patient 1 Injuries

26yo Male

- Alert, calling for help
- Left arm traumatic amputation at elbow with active arterial bleeding
- Sign of penetrating injuries on chest

Patient 2 Injuries

21yo Female

- Reduced consciousness, agitated
- Facial trauma and signs of head injury
- Compromised airway

Both patients in smoke filled compartment with minimal flooding
Immediate CUT is being performed by personnel wearing IRF[1] gear

(1) Immediale Response Firefighting (IRF): Normal uniform, flash gear, light breathing apparatus

Patient - CUT Critical Actions

- Check scene safety
- There is no imminent threat to you or your unit at that moment
- Assess for Massive Hemorrhage
- Remove patient from compartment to safe space
- Address hypothermia

TCCC in the Maritime Scenario 7

Phase 1: Care Under Threat at Sea (CUT)

Patient 1 Injuries

26yo Male

- Alert, calling for help
- Left arm traumatic amputation at elbow with active arterial bleeding
- Sign of penetrating injuries on chest

Patient 2 Injuries

21yo Female

- Reduced consciousness, agitated
- Facial trauma and signs of head injury
- Compromised airway

Patient 1-CUT Critical Actions

- ❑ Check scene safety. Use appropriate Personal Protective Equipment (PPE)
- ❑ Apply tourniquet to left arm amputation**
- ❑ Remove patient from smoke filled compartment

Patient 2-CUT Critical Actions

- ❑ Check scene safety. Use appropriate PPE
- ❑ Remove patient from smoke filled compartment

Any action performed on patient in smoke filled environment should be kept at **absolute minimum **

TCCC in the Maritime Scenario 7

Phase 1->2

Movement to safe zone

Prioritize who is moved first

Prioritize treatment resources in safe zone

TCCC in the Maritime Scenario 7

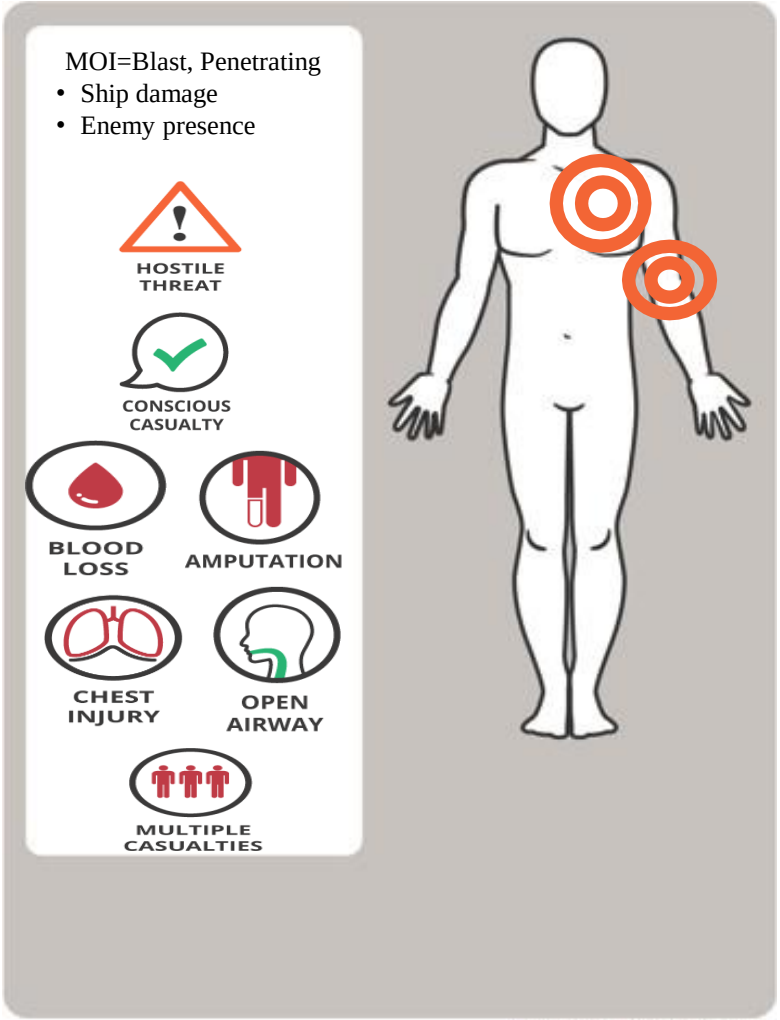
Phase 2: Tactical Field Care (TFC)-Exam

Patient 1

What You See/Feel:

- Alert, screaming for help
- Left arm traumatic amputation at elbow with active arterial bleeding
- Penetrating left anterior chest trauma at rib 3, no posterior chest wall exit wound
- Collapsed left lung without life threatening symptoms (i.e. no tension physiology)

Patient is now moved to safety - Battle Dressing Station



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TCCC in the Maritime Scenario 7

Phase 2: Tactical Field Care (TFC)

Patient 1- TFC Critical Actions

- M:
  - Reassess Tourniquet after patient movement, place 2nd Tourniquet as needed
  - Check for other massive hemorrhage
- A:
  - Assess airway.
    - Airway patent, tolerating secretions
  - Chin-lift/jaw thrust or Naso-Pharyngeal Airway (NPA) as needed
- R:
  - Assessment of breathing.
    - Multiple penetrating fragments Decreased chest wall rise compared to right
  - Needle decompression & application of chest seal
  - Oxygen administration
- C:
  - Recognize hemorrhagic shock
    - Weak pulses in all extremities (BP>80), tachycardic. Skin cold to touch
  - *Intravenous/Intraosseous (IV/IO) access***
  - *Tranexamic Acid (TXA) administration***
  - Blood transfusion/activation of walking blood bank
- H/H:
  - Assessment of mental status (AVPU).
    - Increasingly somnolent, still responding
  - Hypothermia prevention
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with consideration of effect on hemodynamics (Ketamine recommended)
- Antibiotics:
  - *Intramuscular/Intravenous (IM/IV) according to national procedures (no oral meds with Altered Mental Status)***
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

*** If transport to main dressing station is expedited, may consider postponing IV/IO access, TXA, antibiotics ***

TCCC in the Maritime Scenario 7

Phase 2->3

Patient Movement/En Route Care

Patient 1- Movement/ERC Critical Actions

- Patient packaging
  - Properly deploy litter
  - Reassess all interventions after movement onto litter
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention

TCCC in the Maritime Scenario 7

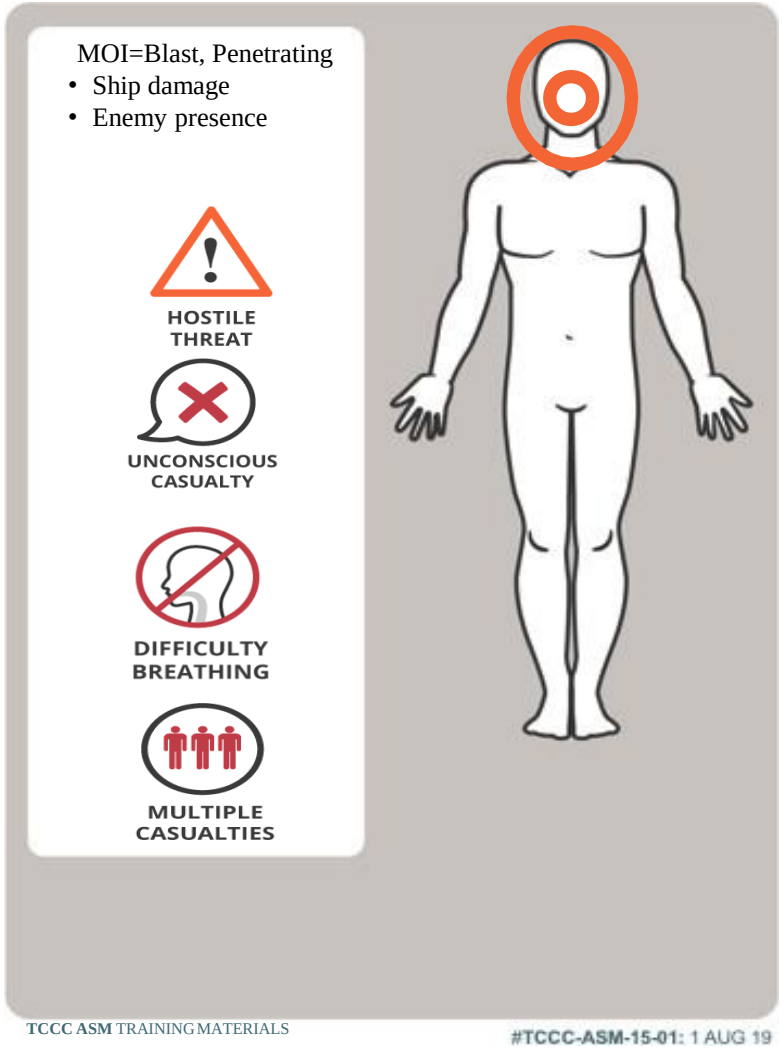
Phase 2: Tactical Field Care (TFC)-Exam

Patient 2

What You See/Feel

- Facial trauma with dislocation and partial loss of jaw
- Loss of multiple teeth and soft tissue loss in lower jaw area
- Closed head injury
- Potential cervical-spine injury

**Patient is now moved to safety
- Battle Dressing Station**



TCCC in the Maritime Scenario 7

Phase 2: Tactical Field Care (TFC)

Patient 2- TFC Critical Actions

- M:
  - Check for massive hemorrhage
- A:
  - Assess airway
  - Chin-lift/jaw thrust, NasoPharyngeal Airway (NPA), as needed
- R:
  - Assessment of breathing.
    - Equal rise and fall of chest
  - Assisted ventilation with Bag Valve Mask as needed
  - Needle decompression & application of chest seal as needed
  - *Oxygen administration*
- C:
  - Pressure dressings/wound packing. Strong palpable pulses in all extremities, tachycardic. Skin warm to touch
  - *Intravenous/Intraosseous (IV/IO) access***
  - *Tranexamic Acid (TXA) administration***
- H/H:
  - Assessment of mental status (AVPU). Altered and agitated
  - Cervical-spine precautions/immobilization for possible cervical-spine injury. CAVE: Airway control
  - Hypothermia prevention
- Pain:
  - Pain medication by Intramuscular/Intravenous/Intraosseous/Intranasal (IM/IV/IO/IN) route with (Ketamine recommended)
- Antibiotics:
  - *Intramuscular/Intravenous (IM/IV) according to national procedures (no PO meds with AMS)***
- Wounds
  - Inspect and dress wounds
  - Look for missed wounds
- Monitoring/Documentation
  - TCCC documentation
  - Vital sign monitoring
  - Frequent reassessment

*** If transport to main dressing station is expedited, may consider postponing IV/IO access, TXA, antibiotics ***

TCCC in the Maritime Scenario 7

Phase 2->3

Patient Movement/En Route Care

Patient 2 – Movement/ERC Critical Actions

- Patient packaging
  - Properly deploy litter. Consider other transport position?
  - C-spine support & Head elevation consideration due to TBI
  - Reassess all interventions after movement onto litter
  - Establish easy access to monitoring equipment and vascular access locations
  - Ensure hypothermia prevention at all times
- Communications
  - Establish communications with main Battle Dressing Station and Damage Control Coordinator
  - Wait for safe route from Damage Control Coordinator
  - Provide MIST report to receiving team if communications established
- Patient care
  - Frequent monitoring/reassessment of patient while transporting
  - Maintain hypothermia prevention
  - Maintain adequate Bag Valve Mask ventilations during transport

TCCC in the Maritime Scenario 7

Phase 3: Care w/in Medical Space

Patient 1 - At Main Battle Dressing Station/Movement (ERC) preparation

- Critical actions
  - Reassess MARCH-PAW
  - Monitor VS: HR 155, BP 80/60, RR 24, Pulse Ox 93%, T 35°C (rectal), GCS 13 (M6E3V4)
  - Reassess all interventions. Perform postponed interventions (IV/IO, TXA, antibiotics)
  - Consider blood transfusion
  - Ensure hypothermia prevention at all times
  - Prepare for higher echelon of care. Litter, hypothermia, PPE

- Communications
  - Maintain communications with main Damage Control Coordinator
  - Communicate need for and requirements for patient movement to higher level of care
  - Plan safe route to point of patient pick-up

TCCC in the Maritime Scenario 7

Phase 3: Care w/in Medical Space

Patient 2 - At Main Battle Dressing Station/Movement (ERC) preparation

- Critical actions
  - Reassess MARCH-PAW
  - Consider definitive airway if not already performed
  - Monitor VS HR 115, BP 155/90, RR 22, Pulse Ox 91%, T 37°C (rectal) GCS 10
  - Reassess all interventions. Perform postponed interventions (IV/IO, TXA, antibiotics)
  - Maintain spinal/cervical immobilization
  - Ensure hypothermia prevention at all times
  - Prepare for movement to higher echelon of care. Litter, hypothermia, PPE

- Communications
  - Maintain communications with main Damage Control Coordinator
  - Communicate need for and requirements for patient movement to higher level of care
  - Plan safe route to point of patient pick-up

TCCC in the Maritime Scenario 7

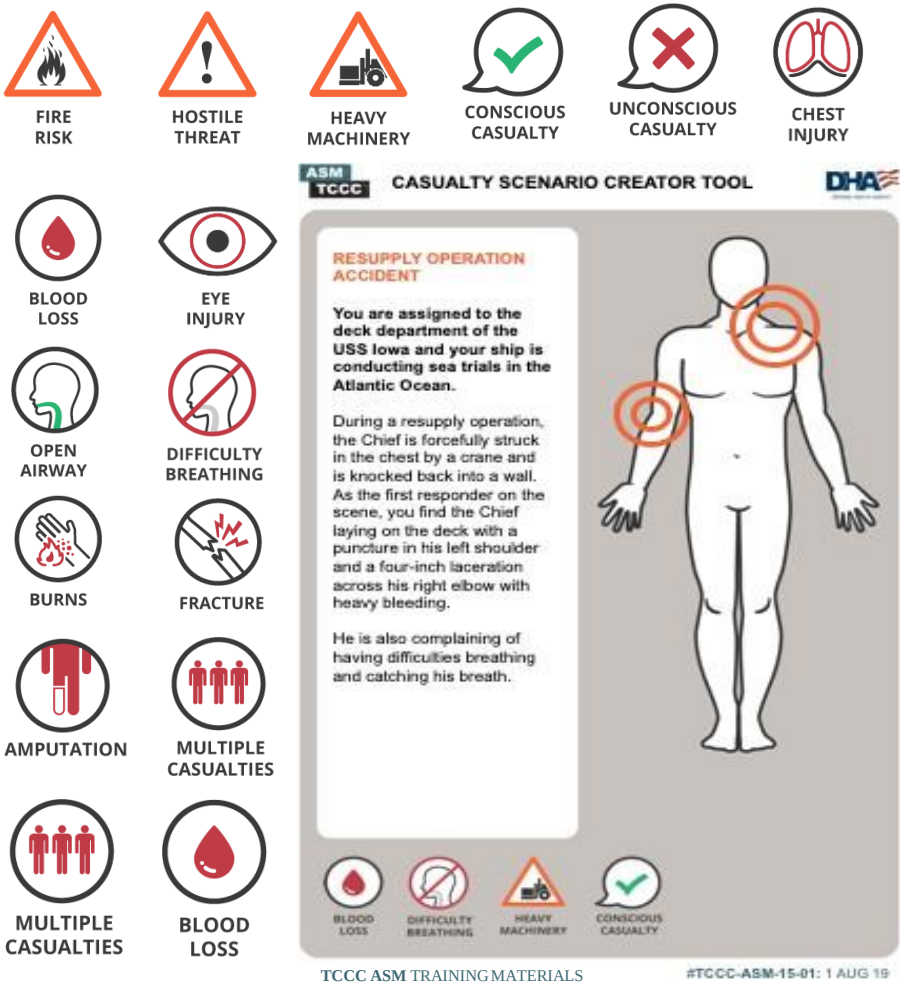
End of Scenario 7

Appendix

CASUALTY SCENARIO CREATOR

Casualty Scenario Creator Tool

The "Casualty Scenario Creator Tool" allows trainers to develop Service-specific scenarios that are customized for their target audience. The PPT slides are fully editable and the various icons below can be dragged and dropped onto the slide for to highlight injury patterns and other details.

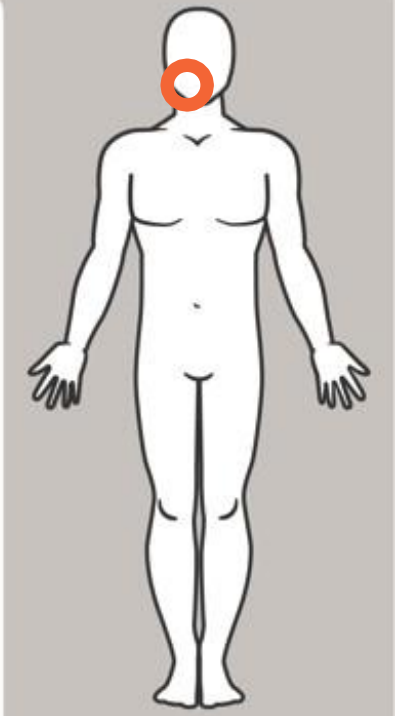


ALL SERVICE MEMBERS **TACTICAL COMBAT CASUALTY CARE**

CASUALTY SCENARIO CREATOR

Insert your scenario text here and drag the corresponding graphics to the figure and bottom:

EYE INJURY



Glossary of Abbreviations

- Airway, Breathing, Circulation, Disability, Exposure **(ABCDE)** Approach
- Altered Mental State **(AMS)**
- Advanced Trauma Life Support **(ATLS)**
- Alert Verbal Pain Unresponsive **(AVPU)**
- Area of Operation **(AO)**
- Assessment of mental status Alert/Verbal/Pain/Unresponsive **(AVPU)**
- Battle Dressings Station **(BDS)**
- Battle Dress Uniform **(BDU)**
- Bag Valve Mask **(BVM)**
- Boarding Party **(BP)**
- Beats per Minute **(BPM)**
- Cervical-Spine **(C-Spine)**
- Casualty Collection Point **(CCP)**
- Combined Task Force **(CTF)**
- Care Under Threat **(CUT)**
- Course of Action **(COA)**
- Damage Control **(DC)**
- Damage Control Coordinator **(DCC)**

Glossary of Abbreviations

continued

- Died of Wounds **(DOW)**
- Guided-Missile Destroyer **(DDG)**
- Dorsalis Pedis **(DP)**
- Fast Attack Craft/Fast Inshore Attack Craft **(FAC/FIAC)**
- Forward **(FWD)**
- Glasgow Coma Scale 15 **(GCS15)**
- Gunshot Wound **(GSW)**
- Heart Rate **(HR)**
- Improvised Explosive Device **(IED)**
- Immediale response firefighting. Normal uniform, flash gear, light breathing apparatus **(IRF)**
- Intramuscular/Intravenous/Intraosseous/Intranasal **(IM/IV/IO/IN)**
- Landing Helicopter Deck **(LHD)**
- Landing Platform/Dock **(LPD)**
- (M) Massive Hemorrhage - Identify and Treat (A) Airway (R) Respiratory System (C) Circulation (H) Head Injury, Hypothermia – (P) Pain (A) Antibiotics (W) Wounds (S) Splinting **(MARCH-PAWS)**
- Massive Hemorrhage **(MH)**
- Mechanism of Injury **(MOI)**
- Medical Evacuation **(MEDEVAC)**
- Medical Naval Panel **(MedNP)**

Glossary of Abbreviations

continued

- (M) Mechanism of Injury (I) Type(s) of Injuries (S) Vital Signs (T) Treatments/Interventions **(MIST)**
- Motor, Eye, Verbal **(M#E#V#)**
- Mother Ship **(MOSHIP)**
- NasoPharyngeal Airway **(NPA)**
- Oxygen **(Ox)**
- (P) Pain (A) Antibiotics (W) Wounds (S) Splinting **(PAWS)**
- (P) Patient/Provider Safety (M) Massive Hemorrhage- Identify and Treat (A) Airway (R) Respiratory System (C) Circulation (H) Head Injury, Hypothermia (P) Patient Evacuation **(P-MARCH-P)**
- Personal Protective Equipment **(PPE)**
- Patient Receiving Unit **(PRU)**
- Pulse Oximetry **(POX)**
- Respiration Rate **(RR)**
- Rigid Hull Inflatable Boat **(RHIB)**
- Tactical Evacuation **(TACEVAC)**
- Tactical Combat Casualty Care **(TCCC)**
- Tactical Field Care **(TFC)**
- Tourniquet **(Tq)**

Glossary of Abbreviations

continued

- Tranexamic Acid **(TXA)**
- Vehicle Borne Improvised Explosive Device **(VBIED)**
- Vessel of Interest **(VOI)**
- Vital Signs **(VS)**
- Walking Blood Bank **(WBB)**
- Water Borne Improvised Explosive Device **(WBIED)**